22/01/25

MODULE 1 : SET, FUNCTIONS & RELATIONS

1. Set
2. Finite set
3. Infinite set
4. Singleton set
5. Empty/void set (null)
6. Equal set
7. Equivalent set ($n(A) = n(B)$)
8. Cardinality
9. subset (proper subset) 2^n-1
10. power set (2^n)

Union
Intersection
Difference
Universal set
Complement

Cross product / cartesian product

Q. If $A = \{5\}$, write the subsets.

subsets of A are $\phi, \{5\}$

No. of subsets $= 2^1 = \underline{\underline{2}}$

Q. $B = \{5, 6\}$, "

$\phi, \{5\}, \{6\}, \{5,6\}$ are subsets of B.

No. of subsets $= 2^2 = \underline{\underline{4}}$

Q. $C = \{5, 6, 7\}$, "

$\phi, \{5\}, \{6\}, \{7\}, \{5,6\}, \{6,7\}, \{5,7\}, \{5,6,7\}$ are subsets of C.

No. of subsets $= 2^3 = \underline{\underline{8}}$

Q. $D = \{5, 6, 7, 8\}$, "

No. of subsets $= 2^4 = 16$

$\phi, \{5\}, \{6\}, \{7\}, \{8\}, \{5,6\}, \{5,7\}, \{5,8\}, \{6,7\}, \{6,8\},$

$\{7,8\}, \{5,6,7\}, \{5,6,8\}, \{5,7,8\}, \{6,7,8\}, \{5,6,7,8\}$

are subsets of D.

Q. $A = \{1, 0\}$, $B = \{2, 4\}$. Find $A \times B$ & $B \times A$.

$A \times B = \{(1,2), (1,4), (0,2), (0,4)\}$

$B \times A = \{(2,1), (2,0), (4,1), (4,0)\}$

Q. $A = \{x, y, z\}$, $B = \{1, 2\}$, $C = \{0, 1\}$.

Find $A \times B \times C$. and $(A \times B) \times C$

$A \times B \times C = \{(x,1,0), (x,1,1), (x,2,0), (x,2,1),$
$(y,1,0), (y,1,1), (y,2,0), (y,2,1),$
$(z,1,0), (z,1,1), (z,2,0), (z,2,1)\}$

$A \times B = \{(x,1), (x,2), (y,1), (y,2), (z,1), (z,2)\}$

$(A \times B) \times C = \{((x,1),0), ((x,1),1), ((x,2),0), ((x,2),1),$
$((y,1),0), ((y,1),1), ((y,2),0), ((y,2),1),$
$((z,1),0), ((z,1),1), ((z,2),0), ((z,2),1)\}$

Q. $A = \{1, 2\}$. Find A^2 and A^3.

Q. If $A = \{a, b, c\}$, $B = \{x, y\}$, $C = \{0, 1\}$. Find

(1) $A \times B \times C$

(2) $C \times A \times B$

(3) $B \times B \times B$

(4) $A \times (B \times C)$

Ans:

$A = \{1, 2\}$.

$A \times A = \{(1,1), (1,2), (2,1), (2,2)\}$

$A \times A \times A = \{(1,1,1), (1,1,2), (1,2,1), (1,2,2),$
$(2,1,1), (2,1,2), (2,2,1), (2,2,2)\}$

Q. $A = \{a, b, c\}$, $B = \{x, y\}$, $C = \{0, 1\}$.

(1) $A \times B \times C = \{(a,x,0), (a,x,1), (a,y,0), (a,y,1),$
$(b,y,0), (b,y,1), (b,x,0), (b,x,1),$
$(c,x,0), (c,x,1), (c,y,0), (c,y,1)\}$

(2) $C \times A \times B = \{(0,a,x), (0,a,y), (0,b,x), (0,b,y),$
$(0,c,x), (0,c,y), (1,a,x), (1,a,y),$
$(1,b,x), (1,b,y), (1,c,x), (1,c,y)\}$

(3) $B \times B \cancel{\times B} = \{(x,x), (x,y), (y,x), (y,y)\}$

$B \times B \times B = \{(x,x,x), (x,x,y), (x,y,x), (x,y,y),$
$(y,x,x), (y,x,y), (y,y,x), (y,y,y)\}$

(4) $A \times (B \times C)$

$B \times C = \{(x,0), (x,1), (y,0), (y,1)\}$

$A \times (B \times C) = \{(a,(x,0)), (a,(x,1)), (a,(y,0)), (a,(y,1)),$
$(b,(x,0)), (b,(x,1)), (b,(y,0)), (b,(y,1)),$
$(c,(x,0)), (c,(x,1)), (c,(y,0)), (c,(y,1))\}$

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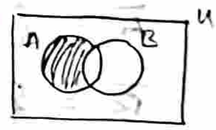
Q. Explain set operations

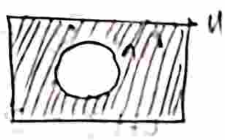
Union : $A \cup B = \{x \mid x \in A \vee x \in B\}$

Intersection : $A \cap B = \{x \mid x \in A \cancel{\vee} \wedge x \in B\}$

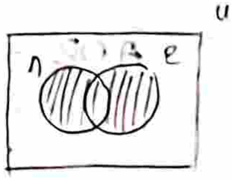


Difference : $A - B = \{x \mid x \in A \wedge x \notin B\}$

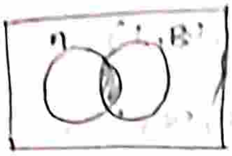
Compliment : $A' = \{x \mid x \in U \wedge x \notin A\}$



Union: set of all the elements in the sets in which the operation is performed.



Intersection - set of all common elements in the set in which the operation is performed.



Q. $A = \{0,2,4,6,8,10\}$, $B = \{0,1,2,3,4,5,6\}$

$C = \{4,5,6,7,8,9,10\}$, $U = \{0,1,2,\dots,15\}$

Find (i) $A \cap B \cap C$

(ii) $A \cup B \cup C$

(iii) $(A \cap B) \cup C$

(iv) $(A \cup B) \cap C$

(v) verify that $\overline{A \cup B} = \overline{A} \cap \overline{B}$ and $\overline{A \cap B} = \overline{A} \cup \overline{B}$ (De-Morgan's Theorem)

Ans: (i) $A \cap B \cap C = \{4,6\}$

(ii) $A \cup B \cup C = \{0,1,2,3,4,5,6,7,8,9,10\}$

(iii) $(A \cap B) = \{0,2,4,6\}$

$(A \cap B) \cup C = \{0,2,4,5,6,7,8,9,10\}$

(iv) $A \cup B = \{0,1,2,3,4,5,6,8,10)$

$(A \cup B) \cap C = \{4,5,6,8,10\}$

(v) $\overline{A \cup B} = \{7,9,11,12,13,14,15\}$ = LHS

$\overline{A} = \{1,3,5,7,9,11,12,13,14,15\}$

$\overline{B} = \{7,8,9,10,11,12,13,14,15\}$

$\overline{A} \cap \overline{B} = \{7,9,11,12,13,14,15\}$ = RHS

LHS = RHS, hence, verified

$\overline{A \cap B} = \{1,3,5,7,8,9,10,11,12,13,14,15\}$ = LHS

$\bar{A} \cup \bar{B} = \{1, 3, 5, 7, 9, 10, 11, 12, 13, 14, 15\} = RHS$

$LHS = RHS$

Hence, verified

Q $A = \{1,2\}$, $B = \{3,4\}$

$A \times B = \{(1,3), (1,4), (2,3), (2,4)\}$

Subsets : $\{\}$, $\{(1,3)\}$, $\{(1,4)\}$, $\{(2,3)\}$, $\{(2,4)\}$, $\{(1,3),(1,4)\}$,
$\{(1,3),(2,3)\}$, $\{(1,3),(2,4)\}$, $\{(1,4),(2,3)\}$, $\{(1,4),(2,4)\}$,
$\{(2,3),(2,4)\}$, $\{(1,3),(1,4),(2,3)\}$,
$\{(1,3),(1,4),(2,4)\}$, $\{(1,3),(2,3),(2,4)\}$,
$\{(1,4),(2,3),(2,4)\}$, $\{(1,3),(1,4),(2,3),(2,4)\}$

These 16 subsets are known as 16 relations.

Q. $A = \{1,2,3\}$. Write $A \times A$. Write subsets of $A \times A$ (Any 5).

$A \times A = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

Subsets $= \{(1,1)\}$, $\{(1,2),(2,1)\}$, $\{\}$, $\{(2,2),(3,2)\}$, $\{(3,3)\}$

No. of subsets $= 2^9$.

NOTE: Reflexive relation: $(x,x) \in R$, $\forall x \in A$

Symmetric relation: If $(a,b) \in R \Rightarrow (b,a) \in R$

Transitive relation : If $(a,b), (b,c) \in R \Rightarrow (a,c) \in R$

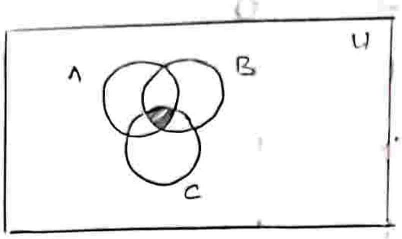
Equivalence relation: A relation which is reflexive, symmetric and transitive.

SET IDENTITIES

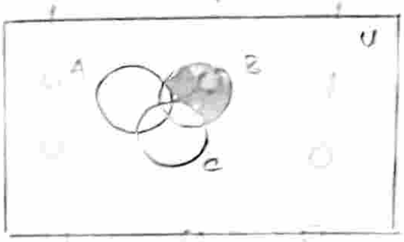
S.No.	Identity	Name
1.	$A \cap U = A$ $A \cup \phi = A$	Identity laws
2.	$A \cup U = U$ $A \cap \phi = \phi$	Domination laws
3.	$A \cup A = A$ $A \cap A = A$	Idempotent laws
4.	$(\overline{\overline{A}}) = A$	complimientation law
5.	$A \cup B = B \cup A$ $A \cap B = B \cap A$	Commutative laws
6.	$A \cup (B \cup C) = (A \cup B) \cup C$ $A \cap (B \cap C) = (A \cap B) \cap C$	Associative laws
7.	$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	Distributive laws
8.	$(\overline{A \cup B}) = \overline{A} \cap \overline{B}$ $(\overline{A \cap B}) = \overline{A} \cup \overline{B}$	De Morgen's laws
9.	$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$	Absorption laws
10.	$A \cup \overline{A} = U$ $A \cap \overline{A} = \phi$	compliment laws

Q. Draw the Venn diagrams of the following sets A, B and C.

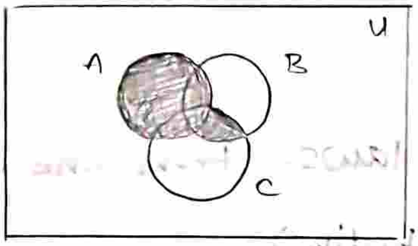
(i) $A \cap B \cap C$



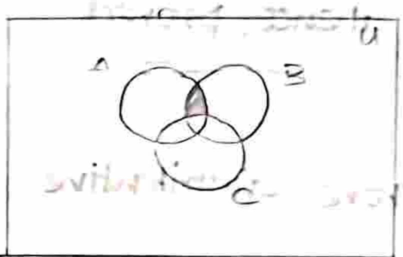
(ii) $B \cap (\overline{A \cup C})$



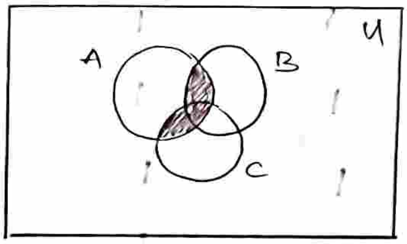
(iii) $A \cup (B \cap C)$



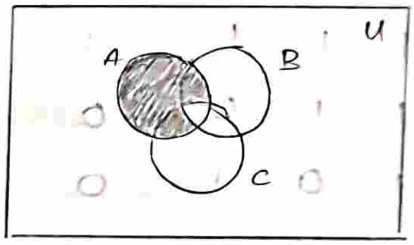
(iv) $A \cap (B - C)$



(v) $(A \cap B) \cup (A \cap C)$



(vi) $(A \cap \overline{B}) \cup (A \cap \overline{C})$



Q. State and prove De-Morgen's law using membership table method.

$(\overline{A \cup B}) = \overline{A} \cap \overline{B}$

If 2 sets $2^2 = 4$ cases

3 sets $2^3 = 8$ cases

A	B	$A \cup B$	$(\overline{A \cup B})$	$\overline{A}$	$\overline{B}$	$\overline{A} \cap \overline{B}$
1	1	1	0	0	0	0
1	0	1	0	0	1	0
0	1	1	0	1	0	0
0	0	0	1	1	1	1

$\hookrightarrow$ LHS $\hookrightarrow$ RHS

Since, LHS = RHS, proved

$\overline{A \cap B} = \overline{A} \cup \overline{B}$

A	B	$A \cap B$	$(\overline{A \cap B})$	$\overline{A}$	$\overline{B}$	$\overline{A} \cup \overline{B}$
1	1	1	0	0	0	0
1	0	0	1	0	1	1
0	1	0	1	1	0	1
0	0	0	1	1	1	1

LHS = RHS

Hence, proved

Q. Prove distributive and associative laws. (Prove one each)

- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$ (distributive)

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
0	1	1	1	1	1	1	1
1	0	0	0	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

LHS = RHS

Hence, proved

$A \cup (B \cup C) = (A \cup B) \cup C$ (Associative)

A	B	C	$B \cup C$	$A \cup (B \cup C)$	$A \cup B$	$(A \cup B) \cup C$
1	1	1	1	1	1	1
1	1	0	1	1	1	1
1	0	1	1	1	1	1
0	1	1	1	1	1	1
1	0	0	0	1	1	1
0	1	0	1	1	1	1
0	0	1	1	1	0	1
0	0	0	0	0	0	0

LHS = RHS

Hence, proved

$|A|$ — cardinality of A

No. of elements

$= 2^n - 1$

PRINCIPLE OF INCLUSION – EXCLUSION

Let A and B be two finite sets, then,

$|A \cup B| = |A| + |B| - |A \cap B|$ $2^2 - 1 = 3$

$2^3 - 1 = 7$

$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$

$$\begin{aligned}|A_1 \cup A_2 \cup A_3 \cup A_4| &= |A_1| + |A_2| + |A_3| + |A_4| - |A_1 \cap A_2| - |A_1 \cap A_3| \\ &\quad - |A_1 \cap A_4| - |A_2 \cap A_3| - |A_2 \cap A_4| - |A_3 \cap A_4| \\ &\quad + |A_1 \cap A_2 \cap A_3| + |A_1 \cap A_2 \cap A_4| + |A_1 \cap A_3 \cap A_4| \\ &\quad + |A_2 \cap A_3 \cap A_4| - |A_1 \cap A_2 \cap A_3 \cap A_4|\end{aligned}$$

Let $A_1, A_2, A_3, A_4, \ldots A_n$ be the finite sets then,

$$|A_1 \cup A_2 \cup A_3 \cdots \cup A_n| = \sum_{1 \le i \le n}^{n} |A_i| - \sum_{1 \le i \le j \le n}^{n} |A_i \cap A_j| + \sum_{1 \le i < j < k \le n}^{n} |A_i \cap A_j \cap A_k| - \cdots + (-1)^{n+1} |A_i \cap A_j \cdots \cap A_n|$$

Q1. In a discrete maths class, every student is a major in cs or maths or both. The no. of students having cs as major is 25, the no. of students having maths as major is 13 and no. of students major in both cs and maths is 8. How many students are in this class?

Ans: $|C| = 25$

$|M| = 13$

$|C \cap M| = \cancel{7}8$

$|C \cup M| = |C| + |M| - |C \cap M|$

$= 25 + 13 - \cancel{7}8$

$= \underline{\underline{30}}$

Q2. How many +ve integers not exceeding 1000 are divisible by 7 or 11.

$|A|$ - No. of elements divisible by 7 = $\left[\frac{1000}{7}\right] = \lfloor 142.85 \rfloor$

$= 142$

$|B|$ = No. of elements divisible by 11 = $\left\lfloor \frac{1000}{11} \right\rfloor = 90$

$|A \cap B|$ = No. of elements divisible by both = $\left\lfloor \frac{1000}{77} \right\rfloor$

$= 12$

$|A \cup B| = |A| + |B| - |A \cap B|$

$= 142 + 90 - 12$

$= \underline{\underline{220}}$

Q. Suppose that there are 1807 freshmen at your school. Of these 453 are taking CS, 567 in maths and 299 in both. How many are not taking either in CS or maths.

$$|A \cup B| = |A| + |B| - |A \cap B|$$
$$= 453 + 567 - 299$$
$$= 1020 - 299 = 721$$

$$|\overline{A \cup B}| = 1807 - 721 = 1086$$

Q. A total of 1232 students have taken Spanish course, 879 French and 114 Russian. Further 103 both Spanish and French, 23 both Spanish and Russian, 14 both French and Russian. If 2092 have taken atleast one of Spanish, French and Russian, How many have taken all Spanish, French & Russian?

$$|S \cup F \cup R| = |S| + |F| + |R| - |S \cap F| - |S \cap R| - |R \cap F| + |R \cap F \cap S|$$

$$2092 = 1232 + 879 + 114 - 103 - 23 - 14 + |S \cap F \cap R|$$

$$2092 = 2111 + 114 - 140 + |S \cap F \cap R|$$

$$\underline{\underline{|S \cap F \cap R| = 7}}$$

Q. Find the number of positive integers not exceeding 1000, that are not divisible by 3, 17 or 35.

$|A|$ = No. divisible by 3 $= \left\lfloor \frac{1000}{3} \right\rfloor = 333$ LCM (3, 17, 35) = 1785

$|B|$ = No. divisible by 17 $= \left\lfloor \frac{1000}{17} \right\rfloor = 58$

$|C|$ = No. divisible by 35 $= \left\lfloor \frac{1000}{35} \right\rfloor = 28$

$|A \cap B \cap C|$ = No. divisible by 3, 17, 35 $= \left\lfloor \frac{1000}{1785} \right\rfloor = 0$

$$|A \cap B| = \left\lfloor \frac{1000}{3 \times 17} \right\rfloor = 19$$

$|B \cap C| = \left\lfloor \frac{1000}{35 \times 17} \right\rfloor = 1$

$|A \cap C| = \left\lfloor \frac{1000}{8 \times 35} \right\rfloor = 9$

$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap C| - |A \cap B| - |B \cap C| + |A \cap B \cap C|$

$= 333 + 58 + 28 - (19 + 1 + 9) + 0$

$= 390$

$|\overline{A \cup B \cup C}| = 1000 - 390$

$= \underline{\underline{610}}$

Q. Find the no. of +ve integers not exceeding 10,000 that are not divisible by 3, 4, 7 or 11.

Ans: $|A| = \left\lfloor \frac{10000}{3} \right\rfloor = 3333$

$|B| = \left\lfloor \frac{10000}{4} \right\rfloor = 2500$

$|C| = \left\lfloor \frac{10000}{7} \right\rfloor = 1428$

$|D| = \left\lfloor \frac{10000}{11} \right\rfloor = 909$

$|A \cup B \cup C \cup D| = |A| + |B| + |C| + |D| - |A \cap B| - |A \cap C| - |A \cap D| - |B \cap C| - |B \cap D| - |C \cap D| + |A \cap B \cap C| + |A \cap B \cap D| + |A \cap C \cap D| + |B \cap C \cap D| - |A \cap B \cap C \cap D|$

$|A \cap B| = \left\lfloor \frac{10000}{3 \times 4} \right\rfloor = 833$

$|A\cap C| = \left\lfloor \frac{10000}{8\times 7} \right\rfloor = 476$

$|A\cap D| = \left\lfloor \frac{10000}{8\times 11} \right\rfloor = 303$

$|B\cap C| = \left\lfloor \frac{10000}{4\times 7} \right\rfloor = 357$

$|B\cap D| = \left\lfloor \frac{10000}{4\times 11} \right\rfloor = 227$

$|C\cap D| = \left\lfloor \frac{10000}{7\times 11} \right\rfloor = 129$

$|A\cap B\cap C| = \lfloor 10000/8\times 4\times 7 \rfloor = 119$

$|A\cap B\cap D| = \lfloor 10000/8\times 4\times 11 \rfloor = 75$

$|A\cap C\cap D| = \lfloor 10000/4\times 7\times 11 \rfloor = 32$

$|A\cap B\cap C\cap D| = \lfloor 10000/8\times 4\times 7\times 11 \rfloor = 10$

$|A\cup B\cup C\cup D| = 8170 - 2325 + 269 - 10$

$= 8439 - 2335$

$= 6104$

$|\overline{A\cup B\cup C\cup D}| = \underline{\underline{3896}}$

Q. Find the no. of the integers not exceeding 100 that are either odd or the square of an integers.

odd, $|A| = 50$

squares, $|B| = 10$

$|A\cap B| = 5$ $(1, 9, 25, 49, 81)$

$$|A \cup B| = |A| + |B| - |A \cap B|$$
$$= 50 + 10 - 5$$
$$= \underline{\underline{55}}$$

Q. How many elements are in $A_1 \cup A_2$. If there are 12 elements in A_1, $|A_2| = 18$ and :

(i) $|A_1 \cap A_2| = 1$ (ii) $|A_1 \cap A_2|$ 6 (iii) $A_1 \cap A_2 = \phi$

(iv) $A_1 \subseteq A_2$

Ans i) $|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2| = 12 + 18 - 1$
$= 29$

ii) $|A_1 \cup A_2| = 12 + 18 - 6 = 24$

iii) $|A_1 \cup A_2| = 12 + 18 - 0 = 30$

iv) $|A_1 \cup A_2| = 12 + 18 - 12 = 18$

Q. Find the no. of +ve integers not exceeding 1000, that are either the square or the cube of an integers.

Square, $|A| = 31$ $\sqrt[2]{1000} = \underline{31}.6$

cube, $|B| = 10$ $\sqrt[3]{1000} = \underline{10}$

$|A \cap B| = 3$ $\sqrt[6]{1000} = \underline{3}.16$

$$|A \cup B \cup C| = 31 + 10 - 3$$
$$= \underline{\underline{38}}$$

EXERCISE QUESTIONS

1. How many elements are in $A_1 \cup A_2$ if there are 12 elements in A_1, 18 elements in A_2, and
   a) $A_1 \cap A_2 = \phi$ b) $|A_1 \cap A_2| = 1$ c) $|A_1 \cap A_2| = 6$
   d) $A_1 \subseteq A_2$

Ans: $|A_1| = 12$ and $|A_2| = 18$

$|A_1 \cup A_2| = |A_1| + |A_2| - |A_1 \cap A_2|$

a) $|A_1 \cap A_2| = 0$

$|A_1 \cup A_2| = 12 + 18 - 0 = \underline{\underline{30}}$

b) $|A_1 \cap A_2| = 1$

$|A_1 \cup A_2| = 30 - 1 = \underline{\underline{29}}$

c) $|A_1 \cap A_2| = 6$

$|A_1 \cup A_2| = 30 - 6 = \underline{\underline{24}}$

d) $|A_1 \cap A_2| = |A_1| = 12$

$|A_1 \cup A_2| = 30 - 12 = \underline{\underline{18}}$

2. There are 345 students at a college who have taken a course in calculus, 212 who have taken a course in discrete mathematics, and 188 who have taken courses in both calculus and discrete mathematics. How many students have taken a course in either calculus or discrete mathematics?

$|C| = 345$

$|M| = 212$

$|C \cap M| = 188$

$|C \cup M| = |C| + |M| - |C \cap M|$

$= 345 + 212 - 188$

$= 345 - 24 = \underline{\underline{369}}$

3. A survey of households in the United States reveals that 96% have at least one television set, 98% have telephone service, and 95% have telephone service and at least one television set. What percentage of households in the United States have neither telephone service nor a television set?

Ans: Consider universet set having 100 elements.

$|TV| = 96$

$|TS| = 98$

$|TV \cap TS| = 95$

$|TV \cup TS| = |TV| + |TS| - |TV \cap TS|$

$= 96 + 98 - 95$

$= 99$

$|\overline{TV \cup TS}| = 100 - 99 = \underline{\underline{1}}.$

4. A marketing report concerning personal computers state that 650,000 owners will buy a printer for their machines next year and 1,250,000 will buy at least one software package. If the report states that 1,450,000 owners will buy either a printer or at least one software package, how many will buy both a printer and at least one software package?

Ans: $|P| = 650,000$

$|S| = 1,250,000$

$|P \cup S| = 1,450,000$

$|P \cup S| = |P| + |S| - |P \cap S|$

$1,450,000 = 650,000 + 1,250,000 - |P \cap S|$

$|P \cap S| = \underline{\underline{4,50,000}}$

5. Find the number of elements in $A_1 \cup A_2 \cup A_3$ if there are 100 elements in each set and if

a) the sets are pairwise disjoint.

b) there are 50 common elements in each pair of sets and no elements in all three sets.

c) there are 50 common elements in each pair of sets and 25 elements in all three sets.

d) the sets are equal.

Ans: a) $|A_1| = |A_2| = |A_3| = 100$

$|A_1 \cap A_2| = |A_1 \cap A_3| = |A_2 \cap A_3| = |A_1 \cap A_2 \cap A_3| = \cancel{\phi} = 0$

$|A_1 \cup A_2 \cup A_3| = 100 + 100 + 100 = \underline{\underline{300}}$

b) $|A_1 \cap A_2| = |A_2 \cap A_3| = |A_1 \cap A_3| = 50$

$|A_1 \cap A_2 \cap A_3| = 0$

$|A_1 \cup A_2 \cup A_3| = 300 - (50+50+50) + 0$

$= \underline{\underline{150}}$

c) $|A_1 \cap A_2| = |A_2 \cap A_3| = |A_1 \cap A_3| = 50$

$|A_1 \cap A_2 \cap A_3| = 25$

$|A_1 \cup A_2 \cup A_3| = 300 - (50+50+50) + 25$

$= 150 + 25 = \underline{\underline{175}}$

d) $|A_1 \cup A_2 \cup A_3| = 300 - (100+100+100) + 100$

$= \underline{\underline{100}}$

6. Find the numbers of elements in $A_1 \cup A_2 \cup A_3$ if there are 100 elements in A_1, 1000 in A_2 and 10,000 in A_3 if

a) $A_1 \subseteq A_2$ and $A_2 \subseteq A_3$.

b) the sets are pairwise disjoint.

c) there are two elements common to each pair of sets and one element in all three sets.

Ans: $|A_1| = 100$

$|A_2| = 1000$

$|A_3| = 10{,}000$

a) $|A_1 \cap A_2| = 100$, $|A_2 \cap A_3| = 1000$

$|A_1 \cap A_3| = 100$

$|A_1 \cap A_2 \cap A_3| = 100$

$|A_1 \cup A_2 \cup A_3| = 100 + 1000 + 10{,}000 - (1000 + 100 + 100) + 100$
$= \underline{\underline{10{,}000}}$

b) $|A_1 \cup A_2 \cup A_3| = 100 + 1000 + 10{,}000$
$= \underline{\underline{11{,}100}}$

c) $|A_1 \cap A_2| = |A_1 \cap A_3| = |A_2 \cap A_3| = 2$

$|A_1 \cap A_2 \cap A_3| = 1$

$|A_1 \cup A_2 \cup A_3| = 100 + 1000 + 10{,}000 - (2+2+2) + 1$
$= \underline{\underline{11{,}095}}$

7. There are 2504 computer science students at a school. Of these, 1876 have taken a course in Java, 999 have taken a course in linux, and 345 have taken a course in C. Further, 876 have taken courses in both Java and linux, 231 have taken courses in both linux and C, and 290 have taken courses in both Java and C. If 189 of these students have taken courses in linux, Java and C, how many of these 2504 students have not taken a course in any of these three subjects?

Ans: $\overset{\text{total}}{\cancel{|J \cup L \cup C|}} = 2504$ $\quad |J \cap L| = 876$

$|J| = 1876$ $\quad |L \cap C| = 231$

$|L| = 999$ $\quad |J \cap C| = 290$

$|C| = 345$ $\quad |J \cap L \cap C| = 189$

$|J \cup L \cup C| = 1876 + 999 + 345 - (876 + 231 + 290) + 189$

$= 3220 - 1397 + 189$

$= 2012$

Required answer $= |\overline{J \cup L \cup C}| = 2504 - 2012$

$= \underline{\underline{492}}$

8. In a survey of 270 college students, it is found that 64 like Brussels sprouts, 94 like broccoli, 58 like cauliflower, 26 like both Brussels sprouts and cauliflower, 22 like both broccoli and cauliflower, and 14 like all three vegetables. How many of the 270 students do not like any of these vegetables?

Ans: Total $= 270$ $\quad |S \cap B| = 26$

$|S| = 64$ $\quad |S \cap C| = 28$

$|B| = 94$ $\quad |B \cap C| = 22$

$|C| = 58$ $\quad |S \cap B \cap C| = 14$

$|S \cup B \cup C| = |S| + |B| + |C| - |S \cap B| - |S \cap C| - |B \cap C| + |S \cap B \cap C|$

$= 64 + 94 + 58 - 26 - 28 - 22 + 14$

$= 154$

$|\overline{S \cup B \cup C}| = 270 - 154$

$= \underline{\underline{116}}$

9. How many students are enrolled in a course either in calculus, discrete mathematics, data structures, or programming languages at a school if there are 507, 292, 312 and 344 students in these courses, respectively; 14 in both calculus and data structures, 213 in both calculus and programming languages; 211 in both discrete mathematics and data structures; 43 in both discrete mathematics and programming languages; and no student may take calculus and discrete mathematics, or data structures and programming languages, concurrently?

Ans: $|C| = 507$ $\quad |C \cap S| = 14$

$|D| = 292$ $\quad |C \cap P| = 213$

$|S| = 312$ $\quad |D \cap S| = 211$

$|P| = 344$ $\quad |D \cap P| = 43$

$|D \cap C| = |S \cap P| = 0$

$$|C \cup D \cup S \cup P| = 507 + 292 + 312 + 344 - (14 + 213 + 211 - 43 + 0)$$

$$= \underline{\underline{974}}$$

10. Find the number of positive integers not exceeding 100 that are not divisible by 5 or 7.

Ans: $|A_1| = \text{No. divisible by } 5 = \left\lfloor \frac{100}{5} \right\rfloor = 20$

$|A_2| = \text{No. divisible by } 7 = \left\lfloor \frac{100}{\cancel{35}\,7} \right\rfloor = \cancel{2}\ 14$

$|A_1 \cap A_2| = \text{No. divisible by both } 5 \,\&\, 7 = \left\lfloor \frac{100}{35} \right\rfloor = 2$

$|A_1 \cup A_2| = 20 + 14 - 2 = 32$

$|\overline{A_1 \cup A_2}| = 100 - 32 = \cancel{35}\ \underline{\underline{68}}$

11. Find the number of positive integers not exceeding 1000 that are not divisible by 3, 17 or 35.

Ans: (Done earlier)

12. Find the number of positive ~~number~~ integers not exceeding 10,000 that are not divisible by 3, 4, 7, or 11.

Ans: $|A_1|$ = No. divisible by 3 = $\left\lfloor \frac{10,000}{3} \right\rfloor = 3333$

$|A_2|$ = No. divisible by 4 = $\left\lfloor \frac{10,000}{4} \right\rfloor = 2500$

$|A_3|$ = No. divisible by 7 = $\left\lfloor \frac{10,000}{7} \right\rfloor = 1428$

$|A_4|$ = No. divisible by 11 = $\left\lfloor \frac{10,000}{11} \right\rfloor = 909$

$|A_1 \cap A_2| = \left\lfloor \frac{10000}{3\times 4} \right\rfloor = 833$

$|A_1 \cap A_3| = \left\lfloor \frac{10000}{3\times 7} \right\rfloor = 476$

$|A_1 \cap A_4| = \left\lfloor \frac{10000}{3\times 11} \right\rfloor = 303$

$|A_2 \cap A_3| = \left\lfloor \frac{10,000}{4\times 7} \right\rfloor = 357$

$|A_2 \cap A_4| = \left\lfloor \frac{10,000}{4\times 11} \right\rfloor = 227$

$|A_3 \cap A_4| = \left\lfloor \frac{10,000}{7\times 11} \right\rfloor = 129$

$|A_1 \cap A_2 \cap A_3| = \left\lfloor \frac{10000}{3\times 4\times 7} \right\rfloor = 119$

$|A_1 \cap A_3 \cap A_4| = \left\lfloor \frac{10000}{3\times 7\times 11} \right\rfloor = 43$

$|A_1 \cap A_2 \cap A_4| = \left\lfloor \frac{10000}{3\times 4\times 11} \right\rfloor = 75$

$|A_2 \cap A_3 \cap A_4| = \left\lfloor \frac{10000}{4\times 7\times 11} \right\rfloor = 32$

$|A_1 \cap A_2 \cap A_3 \cap A_4| = \left\lfloor \frac{10000}{3\times 4\times 7\times 11} \right\rfloor = 10$

$$|A_1 \cup A_2 \cup A_3 \cup A_4| = 3333 + 2500 + 1428 + 909$$
$$- (833 + 476 + 303 + 357 + 227 + 129)$$
$$+ 119 + 43 + 75 + 32 - 10$$

$$= 8170 - 2325 + 259$$

$$= 6104$$

$$|\overline{A_1 \cup A_2 \cup A_3 \cup A_4}| = 10{,}000 - 6104$$
$$= \underline{\underline{3896}}$$

13. Find the number of positive integers not exceeding 100 that are either odd or the square of an integer.

Ans: (Already done)

14. Find the number of positive integers not exceeding 1000 that are either the square or the cube of an integer.

Ans:. (Already done)

18. How many elements are in the union of four sets if each of the sets has 100 elements, each pair of the sets shares 50 elements, each three of the sets share 25 elements, and there are 5 elements in all four sets?

Ans: $|A| = |B| = |C| = |D| = 100$

$|A \cap B| = |A \cap C| = |A \cap D| = |B \cap C| = |B \cap D| = |C \cap D| = 50$

$|A \cap B \cap C| = |B \cap C \cap D| = |A \cap C \cap D| = |A \cap B \cap D| = 25$

$|A \cap B \cap C \cap D| = 5$

$|A \cup B \cup C \cup D| = 4 \times 100 - 6(50) + 4(25) - 5$

$= 400 - 300 + 100 - 5$

$= 200 - 5 = \underline{\underline{195}}$

19.

Ans: $|A| = 50$, $|B| = 60$, $|C| = 70$, $|D| = 80$

$|A \cap B| = |A \cap C| = |A \cap D| = |B \cap C| = |B \cap D| = |C \cap D| = 5$

$|A \cap B \cap C| = |A \cap B \cap D| = |A \cap C \cap D| = |B \cap C \cap D| = 1$

$|A \cap B \cap C \cap D| = 0$

$|A \cup B \cup C \cup D| = 50 + 60 + 70 + 80 - 6(5) + 4 \times 1 - 0$

$= 260 - 30 + 4$

$= \underline{\underline{234}}$

Q0.

Ans: $n = 10$

No. of ~~elements~~ terms in formula $= 2^{10} - 1$

$= 1024 - 1$

$= \underline{\underline{1023}}$

Q1.

Ans: $$|A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5| = \sum_{1 \le i \le 5}^{5} |A_i| - \sum_{1 \le i < j \le 5}^{5} |A_i \cap A_j|$$

$$+ \sum_{1 \le i < j < k \le 5}^{5} |A_i \cap A_j \cap A_k| - \sum_{1 \le i < j < k < l \le 5}^{5} |A_i \cap A_j \cap A_k \cap A_l|$$

$$+ |A_1 \cap A_2 \cap A_3 \cap A_4 \cap A_5|.$$

Q2.

Ans: $$|A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5| = 5 \times 10{,}000 - \overset{1}{\cancel{2}}0 \times 1000 + 10 \times 100 - 5 \times 10 + 1$$

$= 50{,}000 - 10{,}000 + 1000 - 50 + 1$

$= 40{,}000 + 1000 - 50 + 1$

$= 41{,}000 - 50 + 1$

$= \underline{\underline{40{,}951}}$

Q3.

Ans: $$|A_1 \cup A_2 \cup A_3 \cup A_4 \cup A_5 \cup A_6| = \sum_{1 \le i \le 6}^{6} |A_i| - \sum_{1 \le i < j \le 6}^{6} |A_i \cap A_j|$$

25. Let E_1, E_2, and E_3 be three events from a sample space S. Find a formula for the probability of $E_1 \cup E_2 \cup E_3$.

Ans: $$P(E_1 \cup E_2 \cup E_3) = P(E_1) + P(E_2) + P(E_3) - P(E_1 \cap E_2) - P(E_1 \cap E_3) - P(E_2 \cap E_3) + P(E_1 \cap E_2 \cap E_3)$$

Q. Find the no. of positive integers between 1 and 10,000 (inclusive) which are divisible by none of 5, 6, or 8.

Ans: $|A|$ = No. divisible by 5 = $\left\lfloor \frac{10,000}{5} \right\rfloor = 2,000$

$|B|$ = No. divisible by 6 = $\left\lfloor \frac{10,000}{6} \right\rfloor = 1666$

$|C|$ = No. divisible by 8 = $\left\lfloor \frac{10,000}{8} \right\rfloor = 1250$

$2 \,|\, 6, 8$
$3, 4$

$$|A \cap B| = \left\lfloor \frac{10,000}{30} \right\rfloor = 333$$

$$|A \cap C| = \left\lfloor \frac{10,000}{40} \right\rfloor = 250$$

$$|B \cap C| = \left\lfloor \frac{10,000}{24} \right\rfloor = 416$$

$$|A \cap B \cap C| = \left\lfloor \frac{10,000}{120} \right\rfloor = 83$$

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$
$$= 2000 + 1666 + 1250 - (333 + 250 + 416) + 83$$
$$= 4999 - (999)$$
$$= \underline{\underline{4000}}$$

$$|\overline{A \cup B \cup C}| = 10000 - 4000 = \underline{\underline{6000}}$$

05/02/25

Q. For $i = 1, 2, 3, \ldots$ and $A_i = \{i, i+1, i+2, \ldots\}$, then find $\bigcup_{i=1}^{n} A_i$. Also, $\bigcap_{i=1}^{n} A_i$.

$$\bigcup_{i=1}^{n} A_i = A_1 \cup A_2 \cup A_3 \cdots \cup A_n$$

$$\bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap A_3 \cdots \cap A_n.$$

$$\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} \{i, i+1, i+2, \ldots\}$$

$$= \{1, 2, 3, \ldots\} \cup \{2, 3, 4, \ldots\} \cup \{3, 4, 5, \ldots\} \cup \cdots$$
$$\cdots \cup \{n, n+1, n+2, \ldots\}$$

$$= \underline{\{1, 2, 3, \ldots\}} = A_1$$

$$\therefore \bigcap_{i=1}^{n} A_i = A_1 \cap A_2 \cap A_3 \cdots \cap A_n$$

$$= \bigcap_{i=1}^{n} \{i, i+1, i+2, \ldots\}$$

$$= \{1, 2, 3, \ldots\} \cap \{2, 3, 4, \ldots\} \cap \{3, 4, 5, \ldots\} \cap \cdots \cap \{n, n+1, \ldots\}$$

$$= \underline{\{n, n+1, n+2, \ldots\}} = A_n$$

Q. Suppose that $A_i = \{1, 2, 3, 4, 5, \ldots, i\}$, $i = \{1, 2, 3, \ldots\}$.
Find $\bigcup_{i=1}^{\infty} A_i$ and $\bigcap_{i=1}^{\infty} A_i$.

$$\bigcup_{i=1}^{\infty} A_i = \bigcup_{i=1}^{\infty} \{1, 2, 3, 4, \ldots, i\}$$

$$= \{1\} \cup \{1, 2\} \cup \{1, 2, 3\} \cup \cdots \cup \{1, 2, 3, \ldots\}$$

$$= \{1, 2, 3, \ldots\} = \underline{Z^+}$$

$$\bigcap_{i=1}^{\infty} A_i = \bigcap_{i=1}^{\infty} \{1, 2, 3, 4, \ldots, i\}$$

$$= \{1\} \cap \{1, 2\} \cap \{1, 2, 3\} \cap \cdots \cap \{1, 2, 3, \ldots\}$$

$= \{1\} = \underline{A_1}$

Q. Let $A_i = \{\ldots, -2, -1, 0, 1, \ldots, i\}$. Find $\bigcup_{i=1}^{n} A_i$ and $\bigcap_{i=1}^{n} A_i$.

Ans: $\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} \{\ldots, -2, -1, 0, 1, \ldots, i\}$

$= \{\ldots, -2, -1, 0, 1\} \cup \{\ldots, -2, -1, 0, 1, 2\} \cup \{\ldots, -2, -1, 0, 1, 2, 3\}$
$\cup \ldots \cup \{\ldots, -2, -1, 0, 1, 2, \ldots, n\}$

$= \{\ldots, -2, -1, 0, 1, 2, 3, \ldots, n-1, n\}$

$= \underline{A_n}$

$\bigcap_{i=1}^{n} A_i = \bigcap_{i=1}^{n} \{\ldots, -2, -1, 0, 1, \ldots, i\}$

$= \{\ldots, -2, -1, 0, 1\} \cap \{\ldots, -2, -1, 0, 1, 2\} \cap \{\ldots, -2, -1, 0, 1, 2, 3\}$
$\cap \ldots \cap \{\ldots, -2, -1, 0, 1, 2, 3, \ldots, n\}$

$= \{\ldots, -2, -1, 0, 1\} = \underline{A_1}$

Q. State and prove, de Morgan's law.

1) Using a membership table.
2) Set-builder notation and logical equivalence.
3) By showing each side is the subset the other side.

Ans:
1) $(\overline{A \cup B}) = \overline{A} \cap \overline{B}$ and $(\overline{A \cap B}) = \overline{A} \cup \overline{B}$

For $\overline{A \cup B} = \overline{A} \cap \overline{B}$

A	B	$A \cup B$	$\overline{A \cup B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cap \overline{B}$
0	0	0	1	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	1	1	0	0	0	0

LHS = RHS, hence proved

$\overline{A \cap B} = \overline{A} \cup \overline{B}$

A	B	$A \cap B$	$\overline{A \cap B}$	$\overline{A}$	$\overline{B}$	$\overline{A} \cup \overline{B}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

LHS = RHS, hence proved

2) $\overline{A \cup B} = \{x \mid x \notin A \cup B\}$

$= \{x \mid x \notin A \text{ and } x \notin B\}$

$= \{x \mid x \in \overline{A} \text{ and } x \in \overline{B}\}$

$= \{x \mid x \in \overline{A} \cap \overline{B}\} = \overline{A} \cap \overline{B}$

$\overline{A \cup B} = \overline{A} \cap \overline{B}$, LHS = RHS, hence proved.

$\overline{A \cap B} = \{x \mid x \notin A \cap B\}$

$= \{x \mid x \notin A \text{ or } x \notin B\}$

$= \{x \mid x \in \overline{A} \text{ or } x \in \overline{B}\}$

$= \{x \mid x \in \overline{A} \cup \overline{B}\} = \overline{A} \cup \overline{B}$

$\overline{A \cap B} = \overline{A} \cup \overline{B}$, LHS = RHS, hence proved

3)

i) $\overline{A \cup B} = \overline{A} \cap \overline{B}$

$A = B$,
$\Rightarrow A \subseteq B$ and
$B \subseteq A$

$x \in (\overline{A \cup B}) \Rightarrow x \notin \cancel{A \text{ and } x}\ A \cup B$

$\Rightarrow x \notin A \text{ and } x \notin B$

$\Rightarrow x \in \overline{A} \text{ and } x \in \overline{B}$

$\Rightarrow x \in \overline{A} \cap \overline{B}$

$x \in (\overline{A \cup B}) \Rightarrow x \in \overline{A} \cap \overline{B}$

i.e., $\boxed{\overline{A\cup B} \subseteq \bar{A}\cap\bar{B}} \longrightarrow ①$

Again, $x \in (\bar{A}\cap\bar{B}) \Rightarrow x\in\bar{A}$ and $x\in\bar{B}$

$\Rightarrow x\notin A$ and $x\notin B$

$\Rightarrow x\notin (A\cup B)$

$\Rightarrow x\in (\overline{A\cup B})$

$x\in(\bar{A}\cap\bar{B}) \Rightarrow x\in(\overline{A\cup B})$

i.e., $\boxed{\bar{A}\cap\bar{B} \subseteq \overline{A\cup B}} \longrightarrow ②$

From ① and ②, $\boxed{\overline{A\cup B} = \bar{A}\cap\bar{B}}$

ii) $\overline{A\cap B} = \bar{A}\cup\bar{B}$

$x \in (\overline{A\cap B}) \Rightarrow x\notin (A\cap B)$

$\Rightarrow x\notin A$ or $x\notin B$

$\Rightarrow x\in\bar{A}$ or $x\in\bar{B}$

$\Rightarrow x\in\bar{A}\cup\bar{B}$

$x\in(\overline{A\cap B}) \Rightarrow x\in\bar{A}\cup\bar{B}$

i.e., $\boxed{\overline{A\cap B} \subseteq \bar{A}\cup\bar{B}} \longrightarrow ③$

$x\in(\bar{A}\cup\bar{B}) \Rightarrow x\in\bar{A}$ or $x\in\bar{B}$

$\Rightarrow x\notin A$ or $x\notin B$

$\Rightarrow x\notin A\cap B$

$\Rightarrow x\in(\overline{A\cap B})$

$x\in(\bar{A}\cup\bar{B}) \Rightarrow x\in(\overline{A\cap B})$

i.e., $\boxed{\bar{A}\cup\bar{B} \subseteq \overline{A\cap B}} \longrightarrow ④$

From ③ and ④, $\boxed{\overline{A\cap B} = \bar{A}\cup\bar{B}}$

Hence, proved

Q. If A, B and C are sets, then $\boxed{\overline{A\cap B\cap C} = \bar{A}\cup\bar{B}\cup\bar{C}}$.

1) Prove by membership method.

2) Each side is the subset of the other side.
(Subset method).

Ans:

1)

A	B	C	$A\cap B\cap C$	$\overline{A\cap B\cap C}$	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A}\cup\bar{B}\cup\bar{C}$
0	0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0	1
0	1	0	0	1	1	0	1	1
1	0	0	0	1	0	1	1	1
0	1	1	0	1	1	0	0	1
1	0	1	0	1	0	1	0	1
1	1	0	0	1	0	0	1	1
1	1	1	1	0	0	0	0	0

LHS = RHS, hence proved

2) $\overline{A\cap B\cap C} = \bar{A}\cup\bar{B}\cup\bar{C}$

$x \in \overline{A\cap B\cap C} \Rightarrow x \notin A\cap B\cap C$

$\Rightarrow x \notin A$ or $x \notin B$ or $x \notin C$

$\Rightarrow x \in \bar{A}$ or $x \in \bar{B}$ or $x \in \bar{C}$

$\Rightarrow x \in \bar{A}\cup\bar{B}\cup\bar{C}$

$x \in \overline{A\cap B\cap C} \Rightarrow x \in \bar{A}\cup\bar{B}\cup\bar{C}$

i.e., $\boxed{\overline{A\cap B\cap C} \subseteq \bar{A}\cup\bar{B}\cup\bar{C}} \longrightarrow (1)$

$x \in \bar{A}\cup\bar{B}\cup\bar{C} \Rightarrow x \in \bar{A}$ or $x \in \bar{B}$ or $x \in \bar{C}$

$\Rightarrow x \notin A$ or $x \notin B$ or $x \notin C$

$\Rightarrow x \notin A\cap B\cap C$

$\Rightarrow x \in \overline{A\cap B\cap C}$

i.e., $x \in \overline{A} \cup \overline{B} \cup \overline{C} \Rightarrow x \in \overline{A \cap B \cap C}$

i.e., $\boxed{\overline{A} \cup \overline{B} \cup \overline{C} \subseteq \overline{A \cap B \cap C}} \rightarrow$ ②

From ① and ②, $\boxed{\overline{A \cap B \cap C} = \overline{A} \cup \overline{B} \cup \overline{C}}$

Hence, proved

Q. Let A, B and C be sets. then show that $\overline{A \cup (B \cap C)} = (\overline{C} \cup \overline{B}) \cap \overline{A}$

$\overline{A \cup (B \cap C)} = \overline{A} \cap (\overline{B \cap C})$ (using de-morgan's law)

$= \overline{A} \cap (\overline{B} \cup \overline{C})$ (")

$= (\overline{B} \cup \overline{C}) \cap \overline{A}$ (using commutative prop.)

$= (\overline{C} \cup \overline{B}) \cap \overline{A}$ (using commutative property)

LHS = RHS

Hence, proved

Q. Prove that $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ by

1) membership method.

2) rule method.

1)

A	B	C	$B \cup C$	$A \cap (B \cup C)$	$A \cap B$	$A \cap C$	$(A \cap B) \cup (A \cap C)$
0	0	0	0	0	0	0	0
0	0	1	1	0	0	0	0
0	1	0	1	0	0	0	0
1	0	0	0	0	0	0	0
0	1	1	1	0	0	0	0
1	0	1	1	1	0	1	1
1	1	0	1	1	1	0	1
1	1	1	1	1	1	1	1

LHS = RHS

Hence, proved

2. $A \cap (B \cup C) = \{x \mid x \in A \cap (B \cup C)\}$

$= \{x \mid x \in A \text{ and } x \in (B \cup C)\}$

$= \{x \mid x \in A \text{ and } (x \in B \text{ or } x \in C)\}$

$= \{x \mid (x \in A \text{ and } x \in B) \text{ or } (x \in A \text{ and } x \in C)\}$

$= \{x \mid x \in (A \cap B) \text{ or } x \in (A \cap C)\}$

$= \{x \mid x \in (A \cap B) \cup (A \cap C)\}$

$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

LHS = RHS

Hence, proved

06/02/25

RELATIONS

$A = \{1, 2\}$, $B = \{3, 4\}$

$A \times B = \{(1,3), (1,4), (2,3), (2,4)\}$

subset of $A \times B = \phi$, $\{(1,3)\}$, $\{(1,4)\}$, $\{(2,3)\}$, $\{(2,4)\}$,
$\{(1,3),(1,4)\}$, $\{(1,3),(2,3)\}$, $\{(1,3),(2,4)\}$,
$\{(1,4),(2,3)\}$, $\{(1,4),(2,4)\}$, $\{(2,3),(2,4)\}$,
$\{(1,3),(1,4),(2,3)\}$, $\{(1,3),(1,4),(2,4)\}$,
$\{(1,3),(2,3),(2,4)\}$, $\{(1,4),(2,3),(2,4)\}$,
$\{(1,3),(1,4),(2,3),(2,4)\}$

subsets of $A \times B$ is called a relation.

1. REFLEXIVE RELATION : A relation R on a set A is set to be reflexive if aRa, $\forall a \in A$

2. IRREFLEXIVE RELATION: A relation which is not reflexive

3. SYMMETRIC RELATION: A relation R on a set A is said to be symmetric if aRb $\Rightarrow$ bRa, ~~$\forall a,b \in A$~~

4. Otherwise known as asymmetric relation.

5. Antisymmetric relation: A relation R on a set A is said to be antisymmetric if aRb and bRa, then a = b.

6. TRANSITIVE RELATION: aRb, bRc $\Rightarrow$ aRc

7. EQUIVALENCE RELATION: A relation which satisfies reflexive, symmetric and transitive.

Q1. If $A = \{1, 2, 3, 4\}$, give an example of a relation which is

(i) Reflexive, symmetric but not transitive.

(ii) Reflexive, transitive but not symmetric.

(iii) Symmetric, transitive but not reflexive.

Ans:

(i) $R = \{(1,1), (2,2), (3,3), (4,4), (2,3), (3,2), (2,4), (4,2)\}$

(ii) $R = \{(1,1), (2,2), (3,3), (4,4), (4,2), (2,1), (4,1)\}$

(iii) $R = \{(2,1), (1,2), (3,2), (3,1), (1,3), (2,3), (2,2), (1,1), (3,3)\}$

Q2. let R be the relation on the set of real no.s such that aRb iff a−b is an integer. Is R an equivalence relation?

Ans: ~~To check~~ ~~for~~ if R is a ~~reflexive relation~~,

Given, $aRb \Rightarrow a-b$ is an integer.

① Reflexive:– $aRa \Rightarrow a-a=0$, which is an integer.

$\therefore$ it is reflexive.

② Symmetric:– $aRb \Rightarrow a-b$ is an integer

$\Rightarrow b-a$ is also an integer

$\Rightarrow bRa$

$\therefore$ it is symmetric

③ Transitive:– aRb, bRc i.e., $a-b$ is an integer, $b-c$ is an integer.

Then, $a-b = k_1 \longrightarrow ①$

$b-c = k_2 \longrightarrow ②$

$① + ② = a-c = (a-b)+(b-c) = k_1+k_2$

$\therefore aRc$, since sum of two integers is an integer.

Hence, the relation is an equivalence relation.

Q. If R is a relation in the set Z, defined by $R = \{(x,y) \mid x \in Z, y \in Z, x-y$ is divisible by $3\}$. Prove that R is an equivalence relation.

Ans:

① Reflexive: $xRx \Rightarrow x-x=0$, is divisible by 3.

It is a reflexive relation.

② Symmetric: $xRy \Rightarrow \underbrace{x-y}_{=3k}$ is divisible by 3.

$\Rightarrow \underbrace{y-x}_{=-3k}$ is also divisible by 3.

$\Rightarrow yRx$

∴ It is a symmetric relation.

③ Transitive: $xRy, yRz \Rightarrow x-y=3k_1$ is divisible by 3. ①

$y-z=3k_2$ is divisible by 3. ②

① + ② $\Rightarrow (x-y)+(y-z) = 3k_1+3k_2 = 3(k_1+k_2)$, also divisible by 3.

Since, sum of two numbers divisible by 3 is also divisible by 3.

Hence, it is a transitive relation.

Hence, the relation is an equivalence relation.

Q4. Let R be a relation on Z^+ such that aRb if ab is a perfect square. Check whether R is an equivalence relation.

Ans: ① Reflexive :- $aRa \Rightarrow a \times a$, is a perfect square

$\therefore$ It is a reflexive relation.

② Symmetric :- $aRb \Rightarrow ab$ is a perfect square

$\Rightarrow ba$ is also a perfect square

$\Rightarrow bRa$.

$\therefore$ It is a symmetric relation.

③ Transitive :- $aRb, bRc \Rightarrow ab$ is perfect square and bc is a perfect square

$2 \times 2 = 4$

$2 \times 8a =$

$a \times a$

$$ab = k_1^2 \quad \text{—— ①}$$

$$bc = k_2^2 \quad \text{—— ②}$$

① × ② $\quad ab \times bc = k_1^2 \times k_2^2$

$$ac \times b^2 = k_1^2 \times k_2^2$$

$$ac = \left(\frac{k_1 \times k_2}{b}\right)^2$$

Since, $\left(\frac{k_1 k_2}{b}\right)^2$ is a perfect square, ac is also a perfect square.

$\therefore aRb, bRc \Rightarrow aRc$

Hence, it is a transitive relation

Hence, the relation is an equivalence relation.

Q.5. Show that the "divides" relation on the set of +ve integers, Z^+ is not an equivalence relation.

$a|b \Rightarrow a$ divides b means $\frac{b}{a}$.

Ans: ① Reflexive: $aRa \Rightarrow a|a = \frac{a}{a}$.

$\therefore$ It is a reflexive relation.

② Symmetric: $aRb \Rightarrow a|b$

But $bRa \Rightarrow b|a$, not always true.

$\therefore$ It is not symmetric relation.

③ Transitive :- $aRb, bRc \Rightarrow a|b \Rightarrow b = a \times m \longrightarrow ①$

$\Rightarrow b|c \Rightarrow c = b \times n \longrightarrow ②$

Substituting ① in ②

$c = a \times m \times n$

$c = a \times \text{constant}$

$\Rightarrow \underline{aRc}$.

$\therefore$ It is transitive

$\therefore$ the relation is not an equivalence relation.

07/02/25

POSET

↳ partial order SET

∴ Partial order relation

① Reflexive ↳ Z, $aRb \Rightarrow a \leq b$.

② Anti-symmetric

③ Transitive → A set together with a partial order relation.

→ A relation R on a set A is called a partial order relation or partial ordering relation. If R is reflexive, anti-symmetric and transitive.

↳ eg:- consider the set of integers, Z, aRb if $a \leq b$ is a partial order relation.

① Reflexive: $aRa \Rightarrow a \leq a, \forall Z\ a$.

∴ It is reflexive.

② Anti-symm: $aRb, bRa \Rightarrow a \leq b$ and $b \leq a$

$\Rightarrow a = b$ //

∴ It is anti-symmetric

③ Transitive: $aRb, bRc \Rightarrow a \leq b$ and $b \leq c$

$\Rightarrow a \leq c$

$\Rightarrow aRc$

∴ It is transitive

A set A together with a partial order relation is called a partially ordered set or POSET.

The above example is a POSET.

Q1. Determine whether the relation, "divisibility" is a partial order relation on Z^+.

① Reflexive :- $aRa \Rightarrow a|a \Rightarrow \frac{a}{a}$.

$\therefore$ It is reflexive relation.

② Antisymmetric :- $aRb, bRa \Rightarrow a|b = \frac{b}{a}$ and

$b|a = \frac{a}{b}$

$\Rightarrow a = b$

$\therefore$ It is anti-symmetric.

③ Transitive :- $aRb, bRc \Rightarrow b = a \times m \longrightarrow ①$

$c = b \times n \longrightarrow ②$

Substituting ① in ②,

$c = a \times m \times n$

$\Rightarrow c = a \times \text{const}.$

$\Rightarrow a|c \Rightarrow aRc$

$\therefore$ It is transitive relation.

Hence, the "divisibility" relation is a partial order relation on Z^+

Q.2 Let R be a relation on a set, $A = \{1, 2, 3, 4\}$ defined by $R = \{(1,1), (2,2), (3,3), (4,4), (4,3), (4,2), (4,1), (3,2), (3,1)\}$.

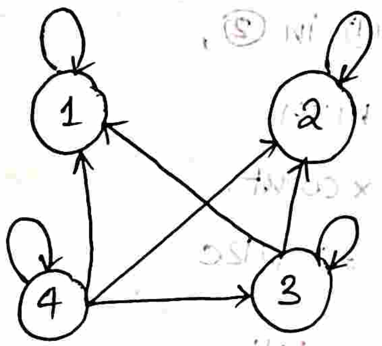
Draw the relation in a Directed graph and also in matrix form.

MATRIX FORM (വരാറില്ല)

$$\begin{array}{c|cccc} & 1 & 2 & 3 & 4 \\ \hline 1 & 1 & 0 & 0 & 0 \\ 2 & 0 & 1 & 0 & 0 \\ 3 & 1 & 1 & 1 & 0 \\ 4 & 1 & 1 & 1 & 1 \end{array}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

DIRECTED GRAPH $\longrightarrow$ digraph (വരാറില്ല)



imp. **HASSE DIAGRAM** (ചോദിക്കും)

POSET can be represented by a diagram known as hasse diagram.

It is a simplified form of digraph.

Since, partial ordering is a reflexive relation, its digraph has loops at all vertices. we need not show these loops since they must be present.

Here, each element is denoted by dot. (•).

Here, the initial vertex is below and have no direction.

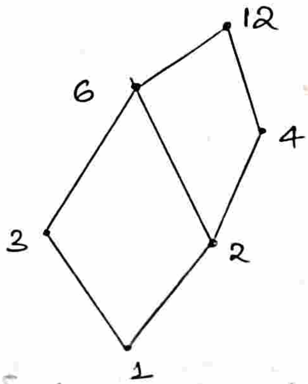
Also, partial ordering is transitive, if there exist (1,2) and (2,3), we need not draw (1,3).

Q. Draw the Hasse Diagram of $(D_{12}, |)$ (divisibility)

D_{12} means divisors of 12.

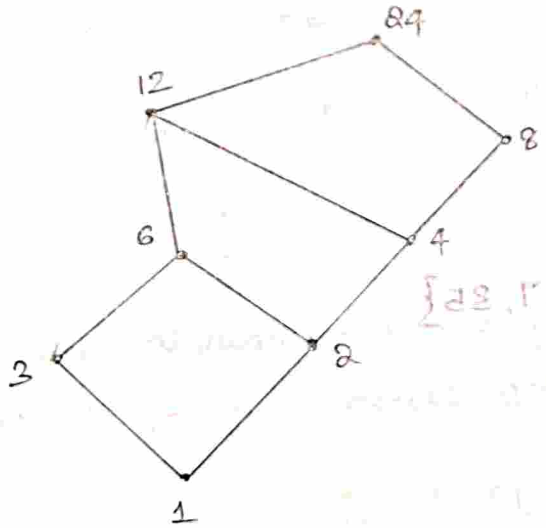
$A = \{1, 2, 3, 4, 6, 12\}$

$R = \{(1,1), (1,2), (1,3), (1,4), (1,6), (1,12),$
$(2,4), (2,2), (2,6), (2,12), (3,3), (3,6),$
$(3,12), (4,4), (4,12), (6,6), (6,12), (12,12)\}$



2. $(D_{24}, |)$

$A = \{1, 2, 3, 4, 6, 8, 12, 24\}$

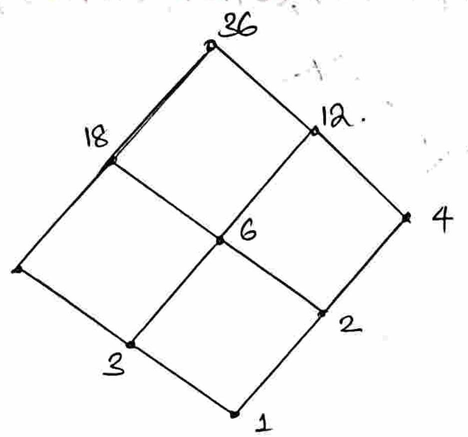


HW.

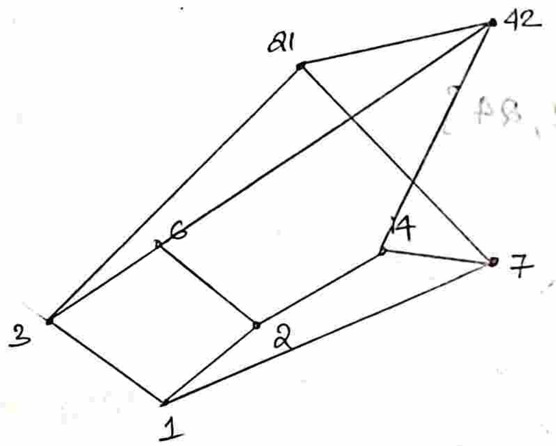
Q. $(D_{36}, |)$, $(D_{42}, |)$, $(D_{35}, |)$, $(D_{100}, |)$

Q. $(D_{36}, |)$

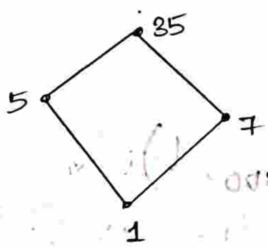
$A = \{1, 2, 3, 4, 6, 9, 12, 18, 36\}$



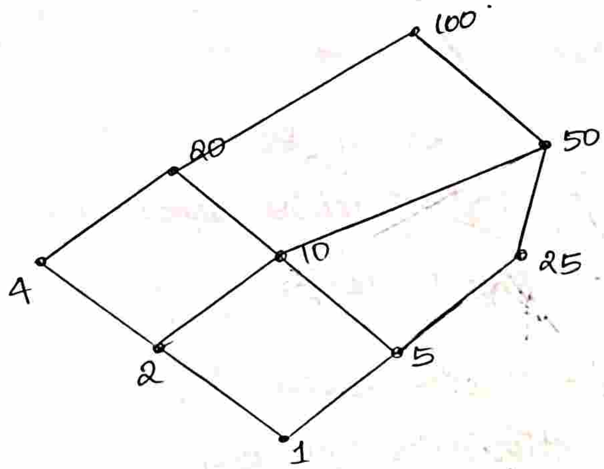
Q. $(D_{42}, |)$ $A = \{1, 2, 3, 6, 7, 14, 21, 42\}$



Q. $(D_{35}, |)$ $A = \{1, 5, 7, 35\}$

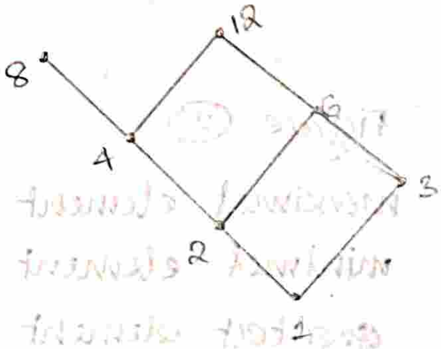


Q. $(D_{100}, |)$ $A = \{1, 2, 4, 5, 10, 20, 25, 50, 100\}$



14/02/25

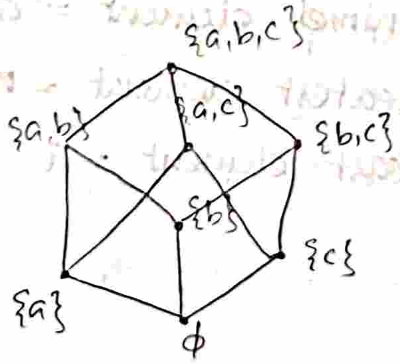
Draw the Hasse diagram representing the partial ordering $\{(a,b) \mid a \text{ divides } b\}$ on $\{1, 2, 3, 4, 6, 8, 12\}$.

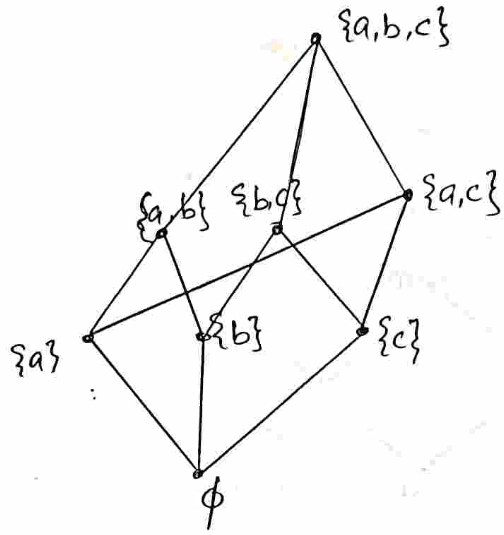


$2^3 = 8$

Q. Draw the Hasse diagram for the partial ordering $\{(A, B) \mid A \subseteq B\}$ on the power set $P(S)$ where $S = \cancel{\{A, B, C\}}$ $\{a, b, c\}$.

$p(S) = \{\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}\}$





greatest & maximal
{a,b,c}
least & minimal
ϕ

1. Maximal element
2. Minimal element
3. Greatest element
4. Least element

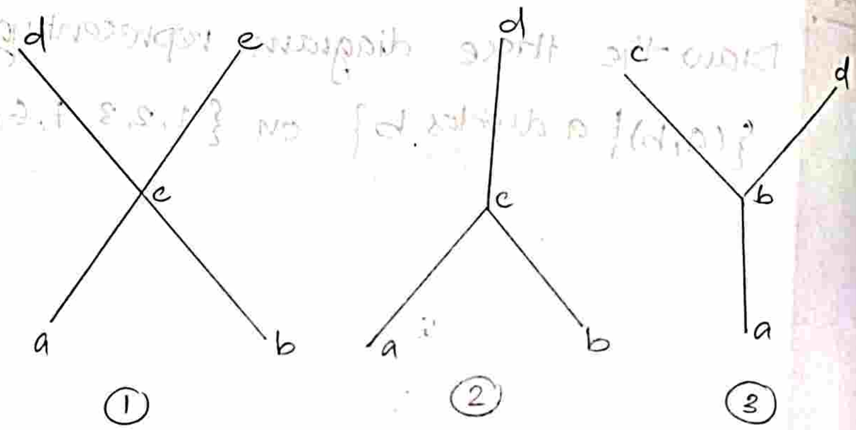


Figure 1
maximal elements = d, e
minimal elements = a, b
greatest element = nil
least element = nil

Figure 2
maximal element = d
minimal element = a, b
greatest element = d
Least element = nil

Figure 3
maximal elements = c, d
minimal element = a
greatest element = nil
least element = a

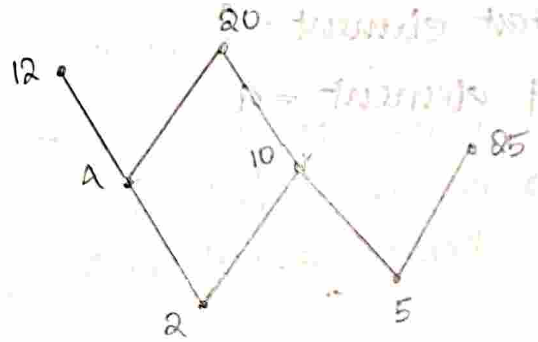
An element of a POSET is called maximel, if it is not less than any element of the POSET, i.e., a is 'maximel' in the POSET $(S, \leq)$ if there is no b element of S such that $a < b$.

An element of a POSET is called minimel, if it is not greater than any element of the POSET, i.e., 'a' is minimel if there is no 'b' element of S such that $b < a$.

A POSET can have more than one maximel element and more than one minimel element.

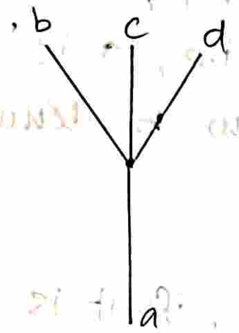
Q. Draw the Hasse diagram of the POSET $\{2,4,5,10,12,20,25\}$ and find the maximel and minimel.

maximel elements = 12, 20, 25
minimel elements = 2, 5



Sometimes there is an element in a POSET that is greater than every other element. Such an element is called the greatest element, i.e., 'a' is the greatest element of the POSET $(S, \leq)$, if $b < a \ \forall \ b \in S$.
Greatest element is unique, then it exists.

An element is called the least element if it is less than all other elements in the POSET $(S, \leq)$, i.e., 'a' is the least element of the POSET $(S, \leq)$, if $a < b \ \forall \ b \in S$.
The least element is unique, then it exists.



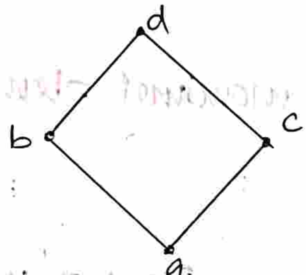
maximel elements = b, c, d

minimel elements = a

greatest element = nil

least element = a

④



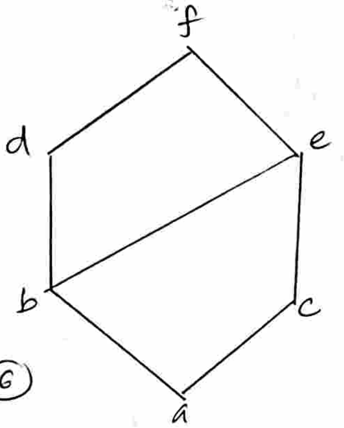
maximel elements = d

minimel elements = a

greatest element = d

least element = a

⑤



maximel element = f

minimel element = a

greatest element = f

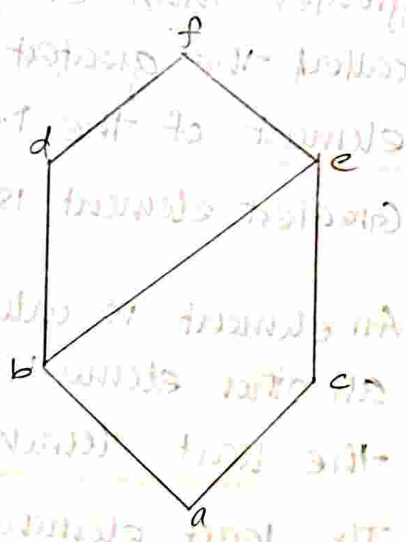
least element = a

⑥

1. <u>UPPER BOUND</u>
2. <u>LEAST UPPER BOUND</u> – LUB – supremum

<u>LOWER BOUND</u>

<u>GREATEST LOWER BOUND</u>



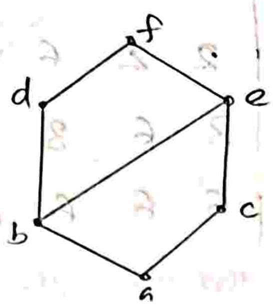
Q. Consider a subset $\{a,b,c\}$.

The upper bound of $\{a,b,c\}$ are e, f. (d is not an upper bound. d have no relation with c).

Here, the least upper bound is e and is denoted by LUB. or supremum.

Let A be a subset of the POSET $(S, \leq)$. If u is an element of S, such that $a \leq u$ $\forall$ elements $a \in A$, then u is called an upper bound of A.

The element α is called LUB of the subset A, if α is an upper bound that is less than every other upper bound of A.



Consider the subset $\{d, e\}$.

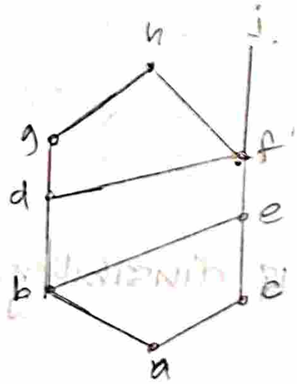
Here, the lower bound of $\{d, e\}$ are a, b. Here, the greatest lower bound is b. → GLB

Or it is known as GLB or infimum.

If l is an element of S, such that $l \leq a$ $\forall$ $a \in A$, then l is called a lower bound of A.

The element y is called the GLB of the subset A, if y is a lower bound of A that is greater than every other lower bound of A.

Q.



Consider the subset $\{a, b, c\}$. Write the upper bound of this subset.

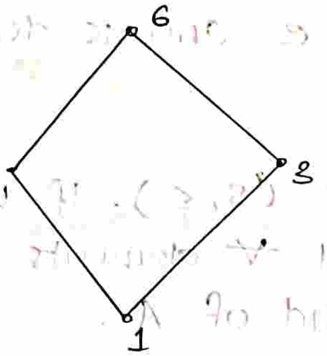
upper bound of $\{a, b, c\}$ $= e, f, h, j$

Consider the subset $\{j, h\}$. Find the lower bounds of $\{j, h\}$.

lower bound of $\{j, h\}$ $= f, e, c, a, b, d$

LATTICE

eg:-



A partially ordered set in which every pair of elements has both an LUB and GLB is called a lattice.

LUB is denoted by join or sum.

$a \vee b$ (join)

$a + b$ (sum)

GLB is denoted by meet or product

$a \wedge b$

$a \cdot b$

LUB

+	1	2	3	6
1	1	2	3	6
2	2	2	6	6
3	3	6	3	6
6	6	6	6	6

of the pair, here, great no is LUB

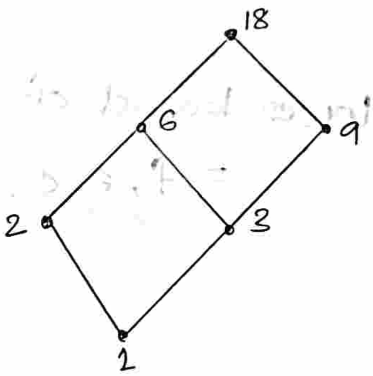
(can avoid (x,x) pairs)

GLB

•	1	2	3	6
1	1	1	1	1
2	1	2	1	2
3	1	1	3	3
6	1	2	3	6

∴ It is a lattice.

Q. let $A = \{1, 2, 3, 6, 9, 18\}$. The relation is divisibility. Draw a Hasse diagram, is it a lattice?



LUB

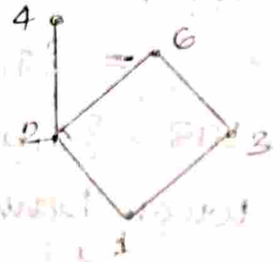
+	1	2	3	6	9	18
1	1	2	3	6	9	18
2	2	2	6	6	18	18
3	3	6	3	6	9	18
6	6	6	6	6	18	18
9	9	18	9	18	9	18
18	18	18	18	18	18	18

GLB

•	1	2	3	6	9	18
1	1	1	1	1	1	1
2	1	2	1	2	1	2
3	1	1	3	3	3	3
6	1	2	3	6	3	6
9	1	1	3	3	9	9
18	1	2	3	6	9	18

∴ It is a lattice.

Q. Draw a Hasse diagram of $A = \{1, 2, 3, 4, 6\}$ and divisibility. Is it a lattice? why?



LUB

+	1	2	3	4	6
1	1	2	3	4	6
2	2	2	6	4	6
3	3	6	3	—	6
4	4	4	—	4	—
6	6	6	6	—	6

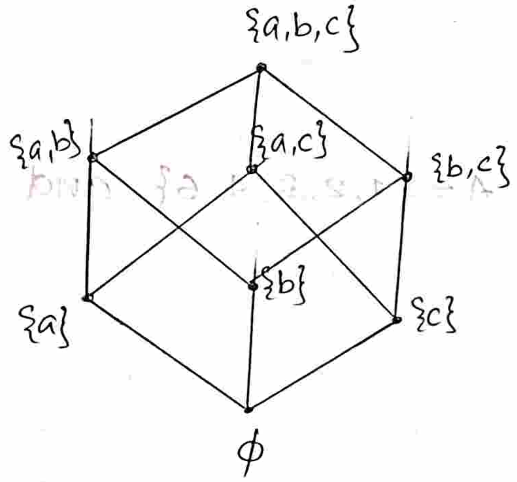
GLB

•	1	2	3	4	6
1	1	1	1	1	1
2	1	2	1	2	2
3	1	1	3	1	3
4	1	2	1	4	2
6	1	2	3	2	6

Here, the pairs (3,4) and (6,4) have no ~~GLB and~~ LUB.
Hence, it is not a lattice.

Q. Let $A = \{a,b,c\}$ and $P(A)$ is its power set. Draw a Hasse diagram $(P(A), \subseteq)$. Find the maximal, minimal, greatest and least element of the POSET. Also, find upper bound, lower bound, LUB, GLB of $\{\{a\},\{b\}\}$. Is it a lattice?

$P(A) = \{\{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}, \phi\}$



Maximal elements = $\{a,b,c\}$

Minimal element = ϕ

Greatest element = $\{a,b,c\}$

Least element = ϕ

Upper bound of $\{\{a\},\{b\}\}$
$= \{a,b\}, \{a,b,c\}$

LUB = $\{a,b\}$

Lower bound of $\{\{a\},\{b\}\}$
$= \phi$

GLB = ϕ

~~ϕ {a} {b} {c} {a,b}~~ {b,c} ~~{a,c}~~ {a,b,c}

∴ This is a lattice

Q1. Let R be the relation on the set of real numbers, such that aRb iff $a-b$ is an integer. Is R an equivalence relation?

Q2. Let M be an integer greater than 1. Show that the relation $\{(a,b) \mid a \equiv b \pmod{m}\}$ is an equivalence relation, on the set of integers.

Q3. Suppose that R is the relation on the set of strings of English letters such that aRb iff $L(a) = L(b)$

where $l(x)$ is the length of the string. Is R an equivalence relation?

Q4. Let R be the relation the set of real numbers such that xRy iff x and y are real numbers that differ by less than 1, i.e., $|x-y|<1$. Check if this is an equivalence relation.

Q5. Show that the greater than or equal to relation is a partial ordering on the set of real numbers.

Q6. Show that the inclusion relation ($\subseteq$) contained is a partial ordering in the power set of a set S.

Q7. Let R be the relation on the set of people such that xRy if x and y are people and x is older than y. Show that R is not an equivalence relation.

Q.8. If $A = \{1,2,3,4,5\}$, $B = \{w,x,y,z\}$, how many elements are in power set of $A \times B$ $[P(A \times B)]$?

Q9. Let A and B be two sets, such that cardinality of B is 3 $[n(B)=3]$. And if there are 4096 relations from A to B, then what is the cardinality of A? $|B|=3$

Q10. Consider the relation on the set of integers and we define $aRb \Rightarrow ab \geq 0$. Check if it is an equivalence relation.

ANSWERS

1. Reflexive :- $aRa \Rightarrow a-a=0$, is an integer!

$\therefore$ R is reflexive

Symmetric :- $aRb \Rightarrow a-b=c$, which is an integer

then $b-a=-c$,

$\Rightarrow bRa$.

$\therefore$ R is symmetric

Transitive :– $aRb, bRc \Rightarrow a-b=k \longrightarrow (1)$
$b-c=\cancel{k}\, l \longrightarrow (2)$

$(1)+(2) \Rightarrow a-c=k+l$ which is an integer

$\Rightarrow aRc$

$\therefore R$ is transitive

Hence, R is an equivalence relation

3. Reflexive : $aRa \Rightarrow l(a)=l(a)$ which is true

$\therefore R$ is reflexive

Symmetric: $aRb \Rightarrow l(a)=l(b)$

Then $l(b)=l(a)$

$\Rightarrow bRa$

$\therefore R$ is ~~asy~~symmetric

Transitive :– $aRb, bRc \Rightarrow l(a)=l(b)$ and $l(b)=l(c)$

Then $l(a)=l(c)$

$\Rightarrow aRc$

$\therefore R$ is transitive

Hence, R is an equivalence relation

4. Reflexive :– $aRa \Rightarrow |a-a|=0<1$

$\therefore R$ is reflexive

Symmetric :– $aRb \Rightarrow |a-b|<1$

Then $|b-a|<1$

$\Rightarrow bRa$ $\therefore R$ is symmetric

Transitive:- aRb, bRc $\Rightarrow |a-b|<1$ and $|b-c|<1$

Then using triangle inequality rule

$$|a-c| \leq |a-b| + |b-c| < 1+1 = 2$$

$$|a-c| < 2$$

$\therefore$ R is not transitive

Hence, R is not an equivalence relation.

5. Reflexive : aRa $\Rightarrow a \geq a$ is true.

$\therefore$ R is reflexive

Anti-symmetric : If aRb, bRa, $a \geq b$ and $b \geq a$.

Then $a=b$.

$\therefore$ R is anti-symmetric

Transitive: aRb, bRc $\Rightarrow a \geq b$ and $b \geq c$

Then $a \geq c$, aRc.

$\therefore$ R is transitive.

Hence greater than or equal to relation is a partial ordering

6. let R be $\subseteq$ inclusion relation.

Reflexive: aRa $\Rightarrow a \subseteq a$, which is true

$\therefore$ R is reflexive

Antisymmetric : aRb, bRa $\Rightarrow a \subseteq b$ and $b \subseteq a$

then $a=b$.

$\therefore$ R is antisymmetric

Transitive: aRb, bRc $\Rightarrow a \subseteq b$ and $b \subseteq c$

Then $a \subseteq c$

$\therefore$ R is transitive

Hence, R is a partial ordering

7 Reflexive: aRa $\Rightarrow$ a is older than a.

Not possible

$\therefore$ R is not reflexive

Symmetric: aRb $\Rightarrow$ a is older than b.

bRa is not true then.

$\therefore$ R is not symmetric

Transitive: aRb, bRc $\Rightarrow$ a is older than b

b is older than c

Then a is older than c too

aRc

$\therefore$ R is transitive

Hence, R is not an equivalence relation

8. No. of elements in $P(A\times B) = 2^{|A|\times|B|}$

$= 2^{5\times 4}$

$= 2^{20}$

$= 1048576$

9. Given $|B| = 3$

No. of relations from A to B $= 2^{|A|\times|B|} = 4096$

$= 2^{|A|\times 3} = 2^{12}$

$\Rightarrow |A|\times 3 = 12$

$|A| = 4$

Hence, A has 4 elements.

10. Reflexive: $aRa \Rightarrow a^2 \geq 0$, since perfect squares greater than or equal to 0.

$\therefore$ R is reflexive

Symmetric: $aRb \Rightarrow ab \geq 0$

Then $ba \geq 0 \Rightarrow bRa$: (commutative)

$\therefore$ R is symmetric

Transitive: $aRb, bRc \Rightarrow ab \geq 0$ and $bc \geq 0$

$ab \geq 0$ means, a and b are either both positive or negative.

lly for $bc \geq 0$, b and c are either both positive or negative.

But if b is 0, then relation is not transitive

$\therefore$ R is not transitive

Hence, R is not an equivalence relation

2. The relation implies $a \equiv b (\text{mod } m) \Rightarrow$ m divides $a-b$.

Reflexive: $aRa \Rightarrow a \equiv a (\text{mod } m) \Rightarrow$ m divides $a-a=0$

since, 0 is divisible by any integer.

$\therefore$ it is reflexive

Symmetric: $aRb \Rightarrow a \equiv b(\text{mod } m) \Rightarrow$ m divides $a-b$

$\Rightarrow a-b=km \longrightarrow ①$

$① \times -1 \Rightarrow -(a-b) = -km$

$b-a = -km$

$\Rightarrow bRa$.

$\therefore$ R is symmetric

Transitive: $aRb, bRc \Rightarrow \underset{①}{a-b=km}$ and $\underset{②}{b-c=lm}$

$① + ② \Rightarrow a-c = m(l+k)$

$\Rightarrow aRc$

$\therefore$ R is transitive $\qquad$ $\therefore$ R is equivalence relation

17/02/25

FUNCTIONS

Let A and B be two non-empty sets.

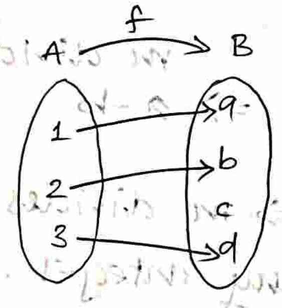
A function or mapping f from A to B, denoted by $f: A \to B$ is a relation from A to B in which every element of A appears exactly once as the first component of an ordered pair in the relation.

If $f: A \to B$ is a fn, then A is called the domain of f, B is called co-domain of f. The set of all images of A under the fn f is called range of f.

eg:- $A = \{1,2,3\}$

$B = \{a,b,c,d\}$

$f: A \to B$ is $f = \{(1,a),(2,b),(3,d)\}$



Here, domain = A

co-domain = B

range = $\{a,b,d\}$

★ All relations are not fns but all fns are relations.

Q. Check the following are fns or not.

1. $\{(x,y) \mid x,y \in \mathbb{R}, y^2 = x\}$

This is not a fn;

when $x = 4$, $y^2 = 4$ $\therefore y = \pm 2$

$\therefore$ Image of 4 is ± 2 i.e., $+2, -2$

i.e., $(4,2), (4,-2)$

$\therefore$ not a function

2. $\{(x,y) \mid x,y \in R,\ y = 3x+1\}$

This is a function. For a single value of x, we get only one value of y.

3. $\{(x,y) \mid x,y \in Q,\ x^2+y^2=1\}$

put $x=0$, $y=\pm 1$

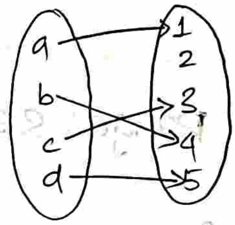
For a single value of x, we get two value of y

$\therefore$ not a fn.

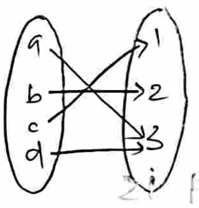
ONE-TO-ONE FUNCTION (INJECTIVE)

A function $f: A \rightarrow B$ is called one-to-one if

$$f(a_1) = f(a_2) \Rightarrow a_1 = a_2$$



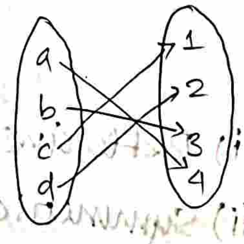
ONTO FUNCTION (SURJECTIVE)



A function $f: A \rightarrow B$ is called onto for all $b \in B$ there is at least one a with $f(a) = b$.

ONE-TO-ONE-ONTO FUNCTION (BIJECTIVE)

A function which is both one-to-one and onto is known as bijective



Q. Check one-to-one and determine the range.

1. $f: Z \rightarrow Z$, $f(x) = 2x+1$

Let $f(x_1) = f(x_2)$

$\Rightarrow 2x_1 + 1 = 2x_2 + 1$

$\Rightarrow 2x_1 = 2x_2$

$\Rightarrow x_1 = x_2$

$\therefore$ it is one-to-one

Range: set of all odd integers.

2. $f: Z \longrightarrow Z,\ f(x) = 2x$

let $f(x_1) = f(x_2)$

$\Rightarrow 2x_1 = 2x_2$

$\Rightarrow x_1 = x_2$

$\therefore$ one-to-one function.

Range: set of all even integers.

3. $f: z \longrightarrow z,\ f(x) = x^3 - x$

Here, $f(0) = 0$

$f(1) = 0$ Here, $0 \neq 1$

$\therefore$ Not one-one

Q. Suppose that the relation R on a set A is represented by the matrix

$$M_R = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix} \text{ is,}$$

(i) Reflexive

(ii) Symmetric

(iii) Anti symmetric

(iv) Transitive

→ The matrix of a relation on a set which a square matrix can be used to determine whether the relation has certain properties.

→ **Reflexive**

$M_{ii} = 1$ or all the main diagonal are equal to 1.

→ **Symmetric**

$M_R = (M_R)^T$ or $M_{ij} = M_{ji}$

→ **Antisymmetric**

$M_{ij} = 1$ and $M_{ji} = 0$, $i \neq j$

→ **Transitive**

$M_{ij} = 1$, $M_{jk} = 1 \Rightarrow M_{ik} = 1$

The above relation is reflexive, symmetric but not antisymmetric because $M_{13} = M_{31} = 0$.

Q. List the ordered pairs in the relation on set $\{1,2,3\}$ corresponding to:

(i) $M_R = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$R = \{(1,1), (1,3), (2,2), (3,1), (3,3)\}$

(ii) $M_R = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$R = \{(1,1), (1,2), (1,3), (2,1), (2,3), (3,3)\}$

Q. Represent the relation of set $\{1,2,3,4\}$ with a matrix

$R = \{(1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}$

Ans: $M_R = \begin{bmatrix} 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

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Q1. Draw the Hasse diagram for divisibility of the set

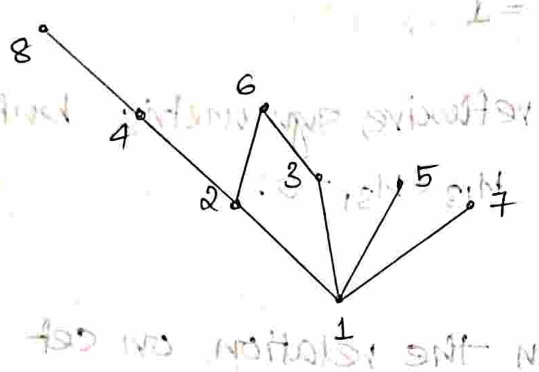
(a) $A = \{1, 2, 3, 4, 5, 6, 7, 8\}$

(b) $B = \{1, 2, 3, 5, 7, 11, 13\}$

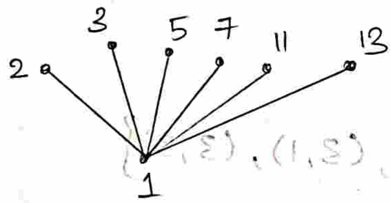
(c) $C = \{1, 2, 3, 6, 12, 24, 36, 48\}$

(d) $D = \{1, 2, 4, 8, 16, 32, 64\}$

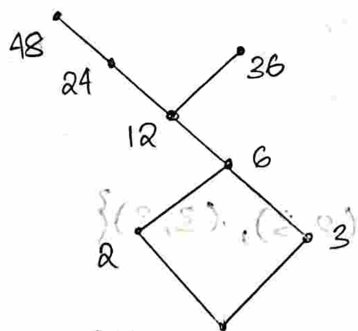
1a)



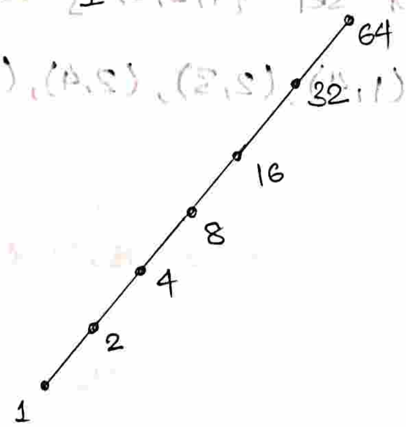
1b)



1c)

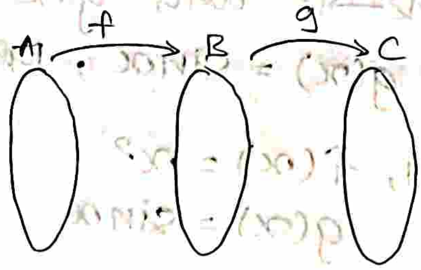


1d)



COMPOSITE FUNCTIONS

If $f: A \rightarrow B$ and $g: B \rightarrow C$ are two functions, then the composite function of f and g from A to C is denoted by $g \circ f$ and is defined as $(g \circ f)(x) = g(f(x))$.



Q. Let $f: R \rightarrow R$ and $g: R \rightarrow R$ defined by $f(x) = x^2$ and $g(x) = x+5$. Find the following:

(1) $f \circ g$ (3) $f \circ f$

(2) $g \circ f$ (4) $g \circ g$

(1) $f(x) = x^2$

$g(x) = x+5$

$(f \circ g)(x) = f(g(x)) = f(x+5)$

$= (x+5)^2$

$= \underline{\underline{x^2 + 10x + 25}}$

(2) $(g \circ f)(x) = g(f(x)) = g(x^2)$

$= \underline{\underline{x^2 + 5}}$

$\boxed{f \circ g \neq g \circ f}$

(3) $(f \circ f)(x) = f(f(x)) = f(x^2)$

$= \underline{\underline{x^4}}$

(4) $(g \circ g)(x) = g(g(x)) = g(x+5)$

$= x+5+5$

$= \underline{\underline{x+10}}$

Q. If $f: R \to R$ and $g: R \to R$ defined by $f(x) = x^2$ and $g(x) = \sin x$, Prove that $f \circ g \neq g \circ f$.

Ans: Given, $f(x) = x^2$

$g(x) = \sin x$

Then, $(f \circ g)(x) = f(g(x))$

$= f(\sin x)$

$= \sin^2 x$

$(g \circ f)(x) = g(f(x))$

$= g(x^2)$

$= \sin x^2$, LHS $\neq$ RHS ($f \circ g \neq g \circ f$)

i.e., $f \circ g \neq g \circ f$

Hence, proved

Q. let $f, g: R \to R$, $f(x) = 2x+5$ and $g(x) = \frac{x-5}{2}$

Find $f \circ g$, $f \circ f$.

Ans: Given, $f(x) = 2x+5$

$g(x) = \frac{x-5}{2}$

$(f \circ g)x = f(g(x))$

$= f\left(\frac{x-5}{2}\right)$

$= 2\left(\frac{x-5}{2}\right) + 5$

$= \underline{x}$

$(f \circ f)x = f(f(x))$

$= f(2x+5)$

$= 4x + 10 + 5$

$= \underline{4x+15}$

Q. Let $f, g, h: \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x+2$, $g(x) = x-2$, and $h(x) = 3x$. Find fog, gof, $fogoh$, $hogof$ and $fo(goh)$

Ans: $f(x) = x+2$
$g(x) = x-2$
$h(x) = 3x$

$(fog)(x) = f(g(x)) = f(x-2)$
$= x-2+2 = \underline{\underline{x}}$

$(gof)(x) = g(f(x)) = g(x+2)$
$= x+2-2 = \underline{\underline{x}}$

$(fogoh)(x) = f(g(h(x))) = f(g(3x))$
$= f(3x-2) = 3x-2+2 = \underline{\underline{3x}}$

$(hogof)(x) = h(g(f(x))) = h(g(x+2))$
$= h(x+2-2) = h(x)$
$= \underline{\underline{3x}}$

$(fo(goh))(x) = f([g(h(x))]) = f([g(3x)])$
$= f(3x-2) = 3x-2+2$
$= \underline{\underline{3x}}$

Q. If f, g, h are functions such that $f(n) = n^2$, $g(n) = n+1$, $h(n) = n-1$, find $fogoh$, $gofoh$, $hofog$.

Ans: $(fogoh)(n) = f(g(h(n))) = f(g(n-1))$
$= f(n+1-1) = f(n)$
$= \underline{\underline{n^2}}$

$(gofoh)(n) = g(f(h(n))) = g(f(n-1))$
$= g((n-1)^2) = (n-1)^2 + 1$
$= n^2 - 2n + 1 + 1 = \underline{\underline{n^2 - 2n + 2}}$

$(h\circ f\circ g)(n) = h(f(g(n))) = h(f(n+1))$

$= h((n+1)^2)$

$= (n+1)^2 - 1$

$= n^2 + 2n + 1 - 1 = \underline{n(n+2)}$

Q ~~Let A =~~ { If f, g, h are functions from $R \longrightarrow R$ defined by $f(x) = x^3 - 4x$; $g(x) = \frac{1}{x^2+1}$, $h(x) = x^4$. Find

(1) $(f\circ g)\circ h$ (2) $f\circ(g\circ h)$ (3) $(h\circ g)\circ f$

Ans: (1) $f\circ g = f(g(x)) = f\left(\frac{1}{x^2+1}\right)$

$= \left(\frac{1}{x^2+1}\right)^3 - 4\left(\frac{1}{x^2+1}\right)$

$((f\circ g)\circ h)(x) = (f\circ g)(h(x))$

$= (f\circ g)(x^4)$

$= \left(\frac{1}{x^8+1}\right)^3 - 4\left(\frac{1}{x^8+1}\right)$

(2) $(g\circ h)(x) = g(h(x)) = g(x^4)$

$= \frac{1}{x^8+1}$

$(f\circ(g\circ h))(x) = \left(f\circ\left(\frac{1}{x^8+1}\right)\right)$

$= \left(\frac{1}{x^8+1}\right)^3 - 4\left(\frac{1}{x^8+1}\right)$

(3) $(h\circ g)(x) = h(g(x))$

$= h\left(\frac{1}{x^2+1}\right) = \left(\frac{1}{x^8+1}\right)^4$

$((h\circ g)\circ f)(x) = h\circ g(f(x))$

$= h\circ g(x^3 - 4x)$

$$= \left(\frac{1}{(x^2+1)^4}\right)^3 - 4\left(\frac{1}{(x^2+1)^4}\right)$$

$$= \left(\frac{1}{x^2+1}\right)^{12} - 4\left(\frac{1}{x^2+1}\right)^4$$

Q. Let $f: R \to R$ and $g: R \to R$ defined by $f(x) = x^2$, $g(x) = x+2$. Find

(1) $f+g$ (2) $f \times g$

(1) $f(x) = x^2$

$g(x) = x+2$

$(f+g)(x) = f(x) + g(x)$

$= x^2 + x + 2$

(2) $(f \times g)(x) = f(x) \cdot g(x)$

$= x^2(x+2) = x^3 + 2x^2$

Q. Let $A = \{1,2,3,4\}$ and R & S be the relation on A defined by $R = \{(1,2), (1,3), (2,4), (4,4)\}$ and $S = \{(1,1), (1,2), (1,3), (2,3), (2,4)\}$. Find:

① SoR ② RoS ③ R^2 ④ R^3

Ans: ① $SoR = \{(1,3), (1,4)\}$

② $RoS = \{(1,2), (1,3), (1,4), (2,4)\}$

③ $R^2 = \{(1,4), (2,4), (4,4)\}$

④ $R^3 = \{(1,4), (2,4), (4,4)\} = RoR^2$

Q. Let $R = \{(1,1), (2,1), (3,2), (4,3)\}$. Find R^2, R^3, R^4.

$R^2 = RoR = \{(1,1), (2,1), (3,1), (4,2)\}$

$R^3 = RoR^2 = \{(1,1), (2,1), (3,1), (4,1)\}$

$R^4 = RoR^3 = \{(1,1), (2,1), (3,1), (4,1)\}$

Q. Let $A = \{1, 2, 3, 4\}$ and $B = \{w, x, y, z\}$ and $C = \{4, 5, 6\}$. $R_1 = \{(1, w), (3, w), (2, x), (4, y)\}$ and $R_2 = \{(w, 5), (x, 6), (y, 4), (y, 6)\}$.
$R_3 = \{(w, 4), (w, 5), (y, 5)\}$.

Determine ① $(R_2 \cup R_3) \circ R_1$
② $(R_2 \circ R_1) \cap (R_3 \circ R_1)$
③ $(R_2 \cap R_3) \circ R_1$

① $R_2 \cup R_3 = \{(w,5), (x,6), (y,4), (y,5), (y,6), (w,4)\}$.
$(R_2 \cup R_3) \circ R_1 = \{(1,5), (1,4), (3,5), (3,4), (2,6), (4,4), (4,5), (4,6)\}$

② $R_2 \circ R_1 = \{(1,5), (3,5), (2,6), (4,4), (4,6)\}$
$R_3 \circ R_1 = \{(1,4), (1,5), (3,4), (3,5), (4,5)\}$
$(R_2 \circ R_1) \cap (R_3 \circ R_1) = \{(1,5), (3,5)\}$

③ $R_2 \cap R_3 = \{(w,5)\}$
$(R_2 \cap R_3) \circ R_1 = \{(1,5), (3,5)\}$

Q1 Let R be the relation on the set of ordered pairs of positive integers such that $((a,b), (c,d)) \in R$ iff $a+d = b+c$. ST is an equivalence relation.

Q2. Determine whether the relation R on the set of all integers is reflexive, symmetric, anti-symmetric, and transitive.

(1) ~~$\{x, y\}$~~ $xy \geq 1$

(2) $x \equiv y \pmod 7$

(3) x is a multiple of y.

Answers.

1) Reflexive: $(a,b)R(a,b) \Rightarrow a+b = b+a$
$= a+b$.

$\therefore$ R is reflexive

Symmetric: $(a,b)R(c,d) \Rightarrow a+d = b+c$ — ①
$(c,d)R(a,b) \Rightarrow$ then $c+b = a+d$, same as equ. ①

$\therefore$ R is symmetric

transitive: $(a,b)R(c,d)$ & $(c,d)R(e,f)$
$\Rightarrow a+d = b+c$ — ① and $c+f = d+e$ — ②
$\Rightarrow d = b+c-a$ — ① and $d = c+f-e$ — ②

Equating ① and ②,

$b+\cancel{c}-a = \cancel{c}+f-e$

$b+e = f+a$

$\Rightarrow a+f = b+e$

$\Rightarrow (a,b)R(e,f)$

$\therefore$ R is transitive

Hence, R is an equivalence relation.

2. (1) $xy \geq 1$

Reflexive: $xRx \Rightarrow x^2 \geq 1$ is not true, since $0^2 = 0$.

$\therefore$ R is not reflexive.

Symmetric: $xRy \Rightarrow xy \geq 1 \Rightarrow yx \geq 1$
$\Rightarrow yRx$.

$\therefore$ R is symmetric.

Anti-symmetric: $xRy \Rightarrow xy \geq 1$

$\Rightarrow yRx$, but $x \neq y$.

$\therefore$ R is not anti-symmetric.

Transitive: $xRy, yRz \Rightarrow xy \geq 1$ and $yz \geq 1$

But xRz need not be true.

For example, if x is +ve and y is +ve

if y and z are -ve.

xz will be -ve.

$\therefore$ not transitive.

(2) Reflexive: $xRx \Rightarrow x-x=0$ is divisible by 7

$\therefore$ R is reflexive

Symmetric: $xRy \Rightarrow x-y = 7k$ — ①

$-1 \times ① \Rightarrow y-x = -7k$

$\Rightarrow yRx$.

$\therefore$ R is symmetric

anti-symmetric: $xRy \Rightarrow x-y = 7k$

$yRx \Rightarrow y-x = -7k$

Then $y=x$ is not always true.

$\therefore$ R is not anti-symmetric

transitive: $xRy, yRz \Rightarrow x-y = k \times 7$ — ①

$y-z = l \times 7$ — ②

$① + ② \Rightarrow x-z = (k+l)7$, $\Rightarrow xRz$

$\therefore$ R is transitive

(3) Reflexive: $xRx \Rightarrow x$ is a multiple of x

$\therefore$ R is reflexive

Symmetric: $xRy \Rightarrow x$ is a multiple of y

$\Rightarrow x = y \times k$ — ① $k \in \mathbb{Z}$ $y = \frac{x}{k}$, $1/k \notin \mathbb{Z}$

But y is not a multiple of x.

$\therefore$ R is not symmetric

Anti symmetric: $xRy \Rightarrow x = ky$ — ①, $k, m \in \mathbb{Z}$

$yRx \Rightarrow y = mx$ — ②

Then, $x = y$

$\therefore$ R is anti-symmetric

Transitive: $xRy, yRz \Rightarrow x = ky$ — ① $k \in \mathbb{Z}$

$y = lz$ — ② $l \in \mathbb{Z}$

Substituting ② in ①,

$x = klz$, $k, l \in \mathbb{Z}$

Then, $x = mz$, here $m = kl \in \mathbb{Z}$

$\therefore$ R is transitive

12/03/25

1. If $A = \{4,5,6,7\}$ and a relation is given by $R = \{(4,4),(5,5),(6,6),(7,7),(4,6),(6,4)\}$. Find the

a) $[4]$ (equivalence class of 4) elements related to 4 (in 2nd position)

b) $[5]$, c) $[6]$ d) $[7]$

$[4] = \{4,6\}$

$[5] = \{5\}$

$[6] = \{6,4\}$

$[7] = \{7\}$

2. i) Define a relation R on $\mathbb{Z}$, by

$R = \{(a,b) \mid a-b$ is divisible by $3\}$

ST R is an equivalence relation.

ii) Determine the equivalence class.

i) Reflexive: $aRa \Rightarrow a-a=0$, is divisible by 3.

$\therefore R$ is reflexive

Symmetric: $aRb \Rightarrow a-b = 3k$ — ①, k is a constant

$b-a = -3k$

$\Rightarrow bRa$

$\therefore R$ is symmetric

Transitive: $aRb, bRc, \Rightarrow a-b=3k$ ① & $b-c=3m$ ②

① + ② $\Rightarrow a-c = 3(k+m)$

$\Rightarrow aRc$

$\therefore R$ is transitive

Hence, R is an equivalence relation.

(ii) $[0] = \{\ldots, -9, -6, -3, 0, 3, 6, 9, 12, \ldots\}$

$[1] = \{\ldots, -8, -5, -2, 1, 4, 7, 10, 13, \ldots\}$

$[2] = \{\ldots, -7, -4, -1, 2, 5, 8, 11, 14, \ldots\}$

$[0] \cup [1] \cup [2] = \{\ldots, -9, -8, -7, \ldots, 0, 1, 2, 3, 4, 5, \ldots\}$

$= \underline{\underline{Z}}$

3. A relation R on Z is defined by xRy if 4 divides $(x-y)$. Check whether it is an equivalence relation. Also, determine the equivalence class.

Ans: Reflexive: $aRa \Rightarrow a-a = 0$, which is divisible by 4.

$\therefore$ R is reflexive.

Symmetric: $aRb \Rightarrow a-b = 4k$

$\Rightarrow b-a = -4k$

$\Rightarrow bRa$.

$\therefore$ R is symmetric.

Transitive: aRb and $bRc \Rightarrow a-b = 4k$ — ①

$b-c = 4m$ — ②

① + ② $\Rightarrow a-c = 4(k+m)$

$\Rightarrow aRc$

$\therefore$ R is transitive

$\therefore$ R is an equivalence relation

$[0] = \{\ldots, -12, -8, -4, 0, 4, 8, 12, \ldots\}$

$[1] = \{\ldots, -11, -7, -3, 1, 5, 9, 13, \ldots\}$

$[2] = \{\ldots, -10, -6, -2, 2, 6, 10, 14, \ldots\}$

$[3] = \{\ldots, -9, -5, -1, 3, 7, 11, 15, \ldots\}$

4. Let f, g, h are functions from $\mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = x^2$, $g(x) = x+5$, $h(x) = \sqrt{x^2+2}$.
Illustrate the associativity property of the composition of three fns using f, g, and h.

Ans: $f\circ(g\circ h) = f[g(h(x))]$

$g(h(x)) = \sqrt{x^2+2} + 5$

$f(g(h(x))) = [\sqrt{x^2+2} + 5]^2$

$= x^2+2+25+10\sqrt{x^2+2}$

$= x^2 + \underline{\underline{10\sqrt{x^2+2}}} + 27$

• $f\circ g \neq g\circ f$ commutativity (usually not equal)

• $f\circ(g\circ h) = (f\circ g)\circ h$ (associativity)

$(f\circ g)\circ h = f(g[h(x)])$

$f\circ g = f(g(x))$
$= (x+5)^2$

$(f\circ g)\circ h = (\sqrt{x^2+2} + 5)^2$

$= x^2+2+25+10\sqrt{x^2+2}$

$= \underline{\underline{x^2+10\sqrt{x^2+2}+27}}$.

Hence, $\underline{\underline{f\circ(g\circ h) = (f\circ g)\circ h}}$

SEMINAR

(a)

Q ST set of +ve odd integers is countable set.

↳ if infinite, should be one-one & onto with natural nos.

$f(n) = 2n-1$

$t \longrightarrow$ odd +ve int

$f(a_1) = f(a_2)$

$t = 2k-1 = f(k)$

$2a_1 - 1 = 2a_2 - 1$

onto

$a_1 = a_2$

one-one

set — infinite — countable only if one to one and onto with natural nos.

/ finite | all countable

(24)

Q. ST set of all integers is countable set.

$$f(n) = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -(n-1)/2 & \text{if } n \text{ is odd} \end{cases}$$

$f(n) = f(m)$

$\frac{n}{2} = \frac{m}{2}$

$n = m$

$-\frac{(n-1)}{2} = -\frac{(m-1)}{2} \Rightarrow m = n.$

(89) Q. ST set of +ve rational numbers is countable set.

sequential form

$\frac{p}{q}$, where $q \neq 0$

diagonal traversal.

$q=1$	$\frac{1}{1}$	$\frac{2}{1}$	$\frac{3}{1}$	$\frac{4}{1}$
$q=2$	$\frac{1}{2}$	$\frac{2}{2}$	$\frac{3}{2}$	$\frac{4}{2}$
$q=3$	$\frac{1}{3}$	$\frac{2}{3}$	$\frac{3}{3}$	$\frac{4}{3}$
$q=4$	$\frac{1}{4}$	$\frac{2}{4}$	$\frac{3}{4}$	$\frac{4}{4}$

$1, \frac{1}{2}, 2, 3, \frac{1}{3}, \ldots$

(89) Q. ST set of real no.s in uncountable

cantor diagonalisation.

Assume real no.s are countable. — if countable, subset is also countable.

$(0,1)$ — Assume 5 elements b/w 0,1

$\{x_1, x_2, x_3, x_4, x_5\}$

$x_1 = 0.123456 \ldots$

$x_2 = 0.246969$

$x_3 = 0.666666 \ldots$

$x_4 = 0.34796 \ldots$

$x_5 = 0.11111 \ldots$

$x = 0.25702 \ldots$

1001

INCLUSION-EXCLUSION PRINCIPLE (app.)

Q6. How many solutions does the equation $x_1+x_2+x_3=11$ have, where x_1, x_2 and x_3 are non-negative integers with $x_1 \leq 3$, $x_2 \leq 4$ and $x_3 \leq 6$?

$\because$ $x_1+x_2+x_3=11$

$x_1 \in \{0,1,2,3\}$, $x_2 \in \{0,1,2,3,4\}$

$x_3 \in \{0,1,2,3,4,5,6\}$

possible solution : { ~~(?)~~, (1,4,6), (2,4,5), (2,3,6),
(3,2,6), (3,3,5), (3,4,4), }

n = no. of variables $= 3$

$r = 11$ = RHS

N = total no. of solutions

$= \cancel{Mt}\ (n+r-1)C_r$

$= (3+11-1)C_{11}$

$= 13C_{11} = \dfrac{13!}{11!\,2!}$ $\dfrac{13\times\cancel{12}^{6}}{}$

$= \underline{78}$

$N(P_1)$ = no. of solutions with $x_1 \geq 4$

$n = 3$

$r = 11-4 = 7$

$= (n+r-1)C_r$

$= (3+7-1)C_7 = 9C_7$

$= \dfrac{9!}{7!\,2!} = \dfrac{9\times 8}{2} = \underline{36}$

$N(P_2)$ = no. of solutions with $x_2 \geq 5$

$n = 3$

$r = 11-5 = 6$

$= (n+r-1)C_r$

$= (3+6-1)C_6 = 8C_6$

$= \dfrac{8!}{6!\,2!} = \dfrac{8\times 7}{2} = \underline{28}$

$N(P_3)$ = no. of solutions with $x_3 \geq 7$

$n = 3$

$r = 11-7 = 4$

$= (n+r-1)C_r$

$= (3+4-1)C_4 = 6C_4$

$= \dfrac{6!}{4!\,2!} = \underline{15}$

$N(P_1P_2)$ = no. of solutions with $x_1 \geq 4$ and $x_2 \geq 5$

$= (n+r-1)C_r$

$n = 3$
$r = 11-(4+5)$
$= 2$

$= (3+2-1)C_2 = 4C_2$

$= \frac{4!}{2!\,2!} = \frac{4 \times 3}{2} = \underline{\underline{6}}$.

$N(P_1P_3)$ = no. of solutions with $x_1 \geq 4$ and $x_3 \geq 7$

$= (n+r-1)C_r$

$= (3+0-1)C_0 = 2C_0$

$r = 11-(4+7)$

$= \frac{2!}{0!\,2!} = \underline{\underline{1}}$

$N(P_2P_3)$ = no. of solutions with $x_2 \geq 5$ and $x_3 \geq 7$

$= (n+r-1)C_r = \underline{0}$

$r = 11-(5+7)$
$= -1$

Since, $5+7 > 11$, $\underline{\underline{N(P_2P_3) = 0}}$.

$N(P_1P_2P_3)$ = no. of soln. with $x_1 \geq 4$, $x_2 \geq 5$ and $x_3 \geq 7$

$= 0$.

Since, $5+4+7 > 11$, $N(P_1P_2P_3) = \underline{0}$.

No. of solutions $= N - (N(P_1) + N(P_2) + N(P_3))$

$+ (N(P_1P_2) + N(P_1P_3) + N(P_2P_3)) - N(P_1P_2P_3)$

$= 78 - (36+28+15) + (6+1+0) - 0$

$= 78 - (79) + 7 = 7-1$

$= \underline{\underline{6}}$

Q7. Find the no. of solutions ~~of~~ the equation $x_1+x_2+x_3+x_4=17$, where x_1, x_2, x_3, x_4 are non-negative integer such that $x_1 \le 3$, $x_2 \le 4$, $x_3 \le 5$, $x_4 \le 8$.

N = total no. of solutions.

$n=4$
$r=17$

$= (n+r-1)C_r$

$= (4+17-1)C_{17} = 20C_{17}$

$= \frac{\cancel{20!}}{\cancel{17!}\,3!} \frac{20 \times 18 \times \cancel{19}}{} = \cancel{1020}\ \underline{1140}$

$N(P_1)$ = total no. of solutions with $x_1 \ge 4$

$r = 17-4$

$= (n+r-1)C_r = (4+13-1)C_{13}$

$= 16C_{13} = \frac{\cancel{16!}}{\cancel{13!}\,3!} \frac{16 \times \cancel{15} \times \cancel{14}}{\cancel{3 \times 2}}$

$= \underline{560}$

$N(P_2)$ = total no. of solutions with $x_2 \ge 5$

$r = 17-5$

$= (n+r-1)C_r = (4+12-1)C_{12}$

$= 15C_{12} = \frac{\cancel{15!}}{\cancel{12!}\,3!} \frac{15 \times \cancel{14} \times 13}{\cancel{3 \times 2}} = 35 \times 13 = \underline{455}$.

$N(P_3)$ = no. of solutions with $x_3 \ge 6$.

$= (n+r-1)C_r = (4+11-1)C_{11}$

$= 14C_{11} = \frac{\cancel{14!}}{\cancel{11!}\,3!} \frac{14 \times 13 \times \cancel{12}}{} = \underline{364}$

$N(P_4)$ = no. of solutions with $x_4 \ge 9$

$= (n+r-1)C_r = (4+8-1)C_8$

$= 11C_8 = \frac{\cancel{11!}}{8!\,3!} \frac{11 \times \cancel{10} \times \cancel{9}}{} = \underline{165}$.

$N(P_1P_2)$ = no. of soln. with $x_1 \geq 4$ and $x_2 \geq 5$.

$= (n+r-1)C_r = (4+8-1)C_8$ $\qquad r = 17-(9) = 8$

$= 11C_8 = \underline{\underline{165}}$.

$N(P_1P_3)$ = no. of soln. with $x_1 \geq 4$ and $x_3 \geq 6$.

$= (n+r-1)C_r = (4+7-1)C_7$ $\qquad r = 17-(10) = 7$

$= 10C_7 = \frac{10!}{7!\,3!} = \frac{10\times 9\times 8}{} = \underline{\underline{120}}$

$N(P_1P_4)$ = no. of soln. with $x_1 \geq 4$ and $x_4 \geq 9$.

$= (n+r-1)C_r = (4+4-1)C_4 = 7C_4$ $\qquad r = 17-13 = 4$.

$= \frac{7!}{4!\,3!} = \frac{7\times 6\times 5}{} = \underline{\underline{35}}$.

$N(P_2P_3)$ = no. of soln. with $x_2 \geq 5$ and $x_3 \geq 6$.

$= (n+r-1)C_r = (4+6-1)C_6 = 9C_6$.

$= \frac{9!}{6!\,3!} = \frac{9\times 8\times 7}{} = \underline{\underline{84}}$.

$N(P_2P_4)$ = no. of soln. with $x_2 \geq 5$ and $x_4 \geq 9$

$= (n+r-1)C_r = (4+3-1)C_3$

Q11. Find the no. of primes less than 100, using the principle of inclusion-exclusion.

N = total no. of elements $= 99$

$\sqrt{100} = 10$

$\therefore$ prime no. not exceeding 10 are $2, 3, 5, 7$.

$N(C_1) = \left\lfloor \frac{100}{2} \right\rfloor = 50$

$N(C_2) = \left\lfloor \frac{100}{3} \right\rfloor = 33$

$N(C_3) = \left\lfloor \frac{100}{5} \right\rfloor = 20$

$N(C_4) = \left\lfloor \frac{100}{7} \right\rfloor = 14$

$N(C_1C_2) = \left\lfloor \frac{100}{3\times 2} \right\rfloor = 16$

$N(C_1C_3) = \left\lfloor \frac{100}{2\times 5} \right\rfloor = 10$

$N(C_1C_4) = \left\lfloor \frac{100}{2\times 7} \right\rfloor = 7$

$N(C_2C_3) = \left\lfloor \frac{100}{3\times 5} \right\rfloor = 6$

$N(C_2C_4) = \left\lfloor \frac{100}{3\times 7} \right\rfloor = 4$

$N(C_3C_4) = \left\lfloor \frac{100}{35} \right\rfloor = 2$

$N(C_1C_2C_3) = \left\lfloor \frac{100}{2\cdot 3\cdot 5} \right\rfloor = 3$

$N(C_2C_3C_4) = \left\lfloor \frac{100}{105} \right\rfloor = 0$

$N(C_1C_3C_4) = \left\lfloor \frac{100}{2\cdot 5\cdot 7} \right\rfloor = 1$

$N(C_1C_2C_4) = \left\lfloor \frac{100}{42} \right\rfloor = 2$

$N(C_1C_2C_3C_4) = \left\lfloor \frac{100}{3\times 2\times 5\times 7} \right\rfloor = \underline{0}$

total no. of primes less than 100 $= N - (N(C_1) + N(C_2) + N(C_3) + N(C_4))$

$+ (N(C_1C_2) + N(C_1C_3) + N(C_1C_4) + N(C_2C_3) + N(C_2C_4) + N(C_3C_4))$

$- (N(C_1C_2C_3) + N(C_1C_2C_4) + N(C_1C_3C_4) + N(C_2C_3C_4)) + N(C_1C_2C_3C_4)$

$= 99 - (50 + 33 + 20 + 14) + (16 + 10 + 7 + 6 + 4 + 2)$

$- (3 + 1 + 2 + 0) + 0$

$= 99 - 117 + 45 - 6 = 21$

Q12. Find the no. of primes less than 40.

$N = 39$

$\sqrt{40} = 6 \ldots$

primes within 0 to 6 is 2, 3, 5.

$N(P_1) = \left\lfloor \frac{40}{2} \right\rfloor = 20$

$N(P_2) = \left\lfloor \frac{40}{3} \right\rfloor = 13$

$N(P_3) = \left\lfloor \frac{40}{5} \right\rfloor = 8$

$N(P_1P_2) = \left\lfloor \frac{40}{6} \right\rfloor = 6$

$N(P_1P_3) = \left\lfloor \frac{40}{10} \right\rfloor = 4$

$N(P_2P_3) = \left\lfloor \frac{40}{15} \right\rfloor = 2$

$N(P_1P_2P_3) = \left\lfloor \frac{40}{30} \right\rfloor = 1$

primes less than 40 (w/o 2, 3, 5) $= N - [N(P_1) + N(P_2) + N(P_3)] + (N(P_1P_2) + N(P_1P_3) + N(P_2P_3)] - N(P_1P_2P_3)$

$= 39 - (20 + 13 + 8) + (6 + 4 + 2) - 1$

$= 39 - 41 + 12 - 1 = 9$

total no. of primes less than 40 $= 9 + 3 = \underline{\underline{12}}$